

Gait analysis by parameter row

37 runs in 10 gait families. Each table row is treated as the corresponding filename run number.

Order: Sin-Sin, piecewise Sin-Sin, ordinary Sin-Square, wide Sin-Square,

fixed-amplitude and varying-amplitude timed Sin-Square, Nested Sin, Slow/Fast, Sin-Fourier, Fourier-Square.

These plots reconstruct commanded motion. ROS bags / force CSVs were not supplied.

Force comparisons, tracking errors and measured spectra must be regenerated from recorded data.

Defaults follow PDF TABLES. Dashed alternatives show conflicts with pasted controller code.

Nested Sin row 2: table center 2500 / direction +1 vs controller 2550 / -1.

Slow/Fast rows: table direction +1 vs controller -1. Sin-Fourier: table center2 2000 vs controller 2500.

Double Sin row 2 uses 0.67 Hz literally. Nested Sin frequencies use exact $\text{speed}/(2\pi)$.

Missing table fields come from the corresponding pasted controller / PDF equations.

Square switching uses ideal S1 feedback in these models. Actual delays and tracking change switching times.

Fourier retains denominator $2n+1$ and 299 terms. Slow/Fast uses 99 terms; series scale is not peak amplitude.

High harmonics exceed the timer Nyquist frequency. Dense targets and sampled commands can differ.

All servo axes and tables use degrees = counts $\times$ 360/4096, from encoder zero. Physical neutral is unverified.

U = Uniform; N = Not; L = Lower amplitude; H = Higher amplitude; R = Regular

Sin (U) | Sin (U)

Original key: DOUBLESIN / PDF p.9 / rows map to run numbers

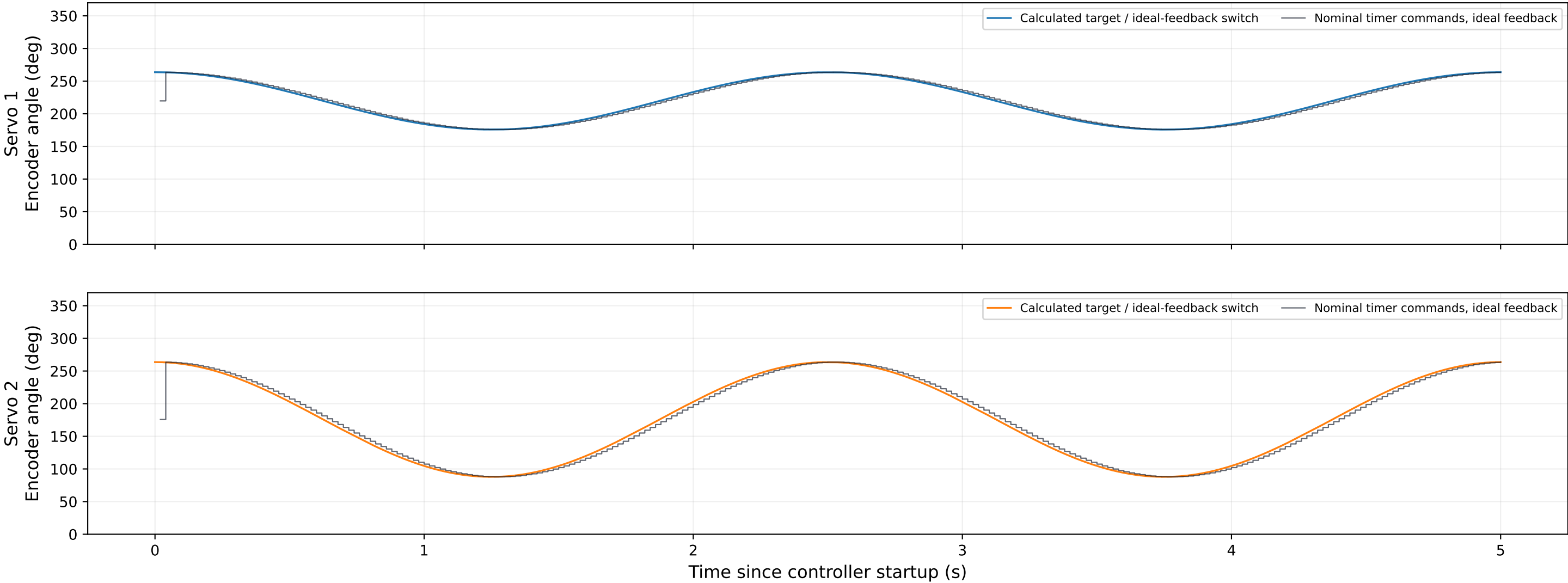
Run	f (Hz)	A1 (deg)	C1 (deg)	A2 / power (deg)	C2 (deg)	Direction	Power ratio	Extra
1	0.4	43.95	219.73	87.89	175.78	1	—	
2	0.67	43.95	219.73	87.89	175.78	1	—	

All angles above are degrees. Centers use encoder zero; amplitudes are offsets. Harmonic amplitudes are scale factors.
Degrees = counts × 360/4096. Raw count parameters are retained in the CSV/JSON and exact position equations.

Sin (U) | Sin (U) / Run 1

f=0.4 Hz; S1 amplitude=43.95°; centers=219.73° / 175.78°; direction=1

MODEL ONLY • Table settings + controller equations • No measured force or motion

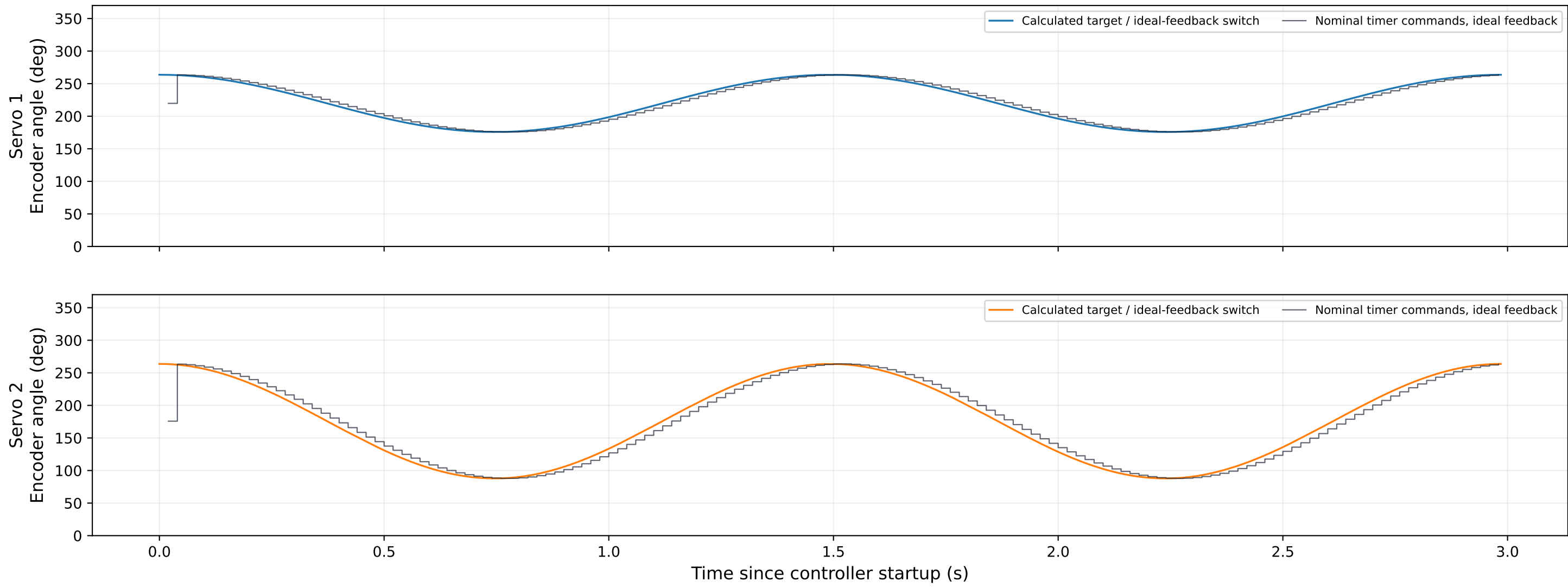


p1=round(2500+500 cos(0.8πt))
p2=round(2000+1000 cos(0.8πt))
Plotted encoder angle = p × 360/4096 degrees. Above position equations retain raw counts for exact rounding.

Sin (U) | Sin (U) / Run 2

f=0.67 Hz; S1 amplitude=43.95°; centers=219.73° / 175.78°; direction=1

MODEL ONLY • Table settings + controller equations • No measured force or motion



p1=round(2500+500 cos(1.34πt))
p2=round(2000+1000 cos(1.34πt))
Plotted encoder angle = p × 360/4096 degrees. Above position equations retain raw counts for exact rounding.
0.67 Hz used literally from table; exact controller omega for this row was not provided.

Sin (LNU) | Sin (LU)

Original key: experimental_sinusoidalYaw / PDF p.2 / rows map to run numbers

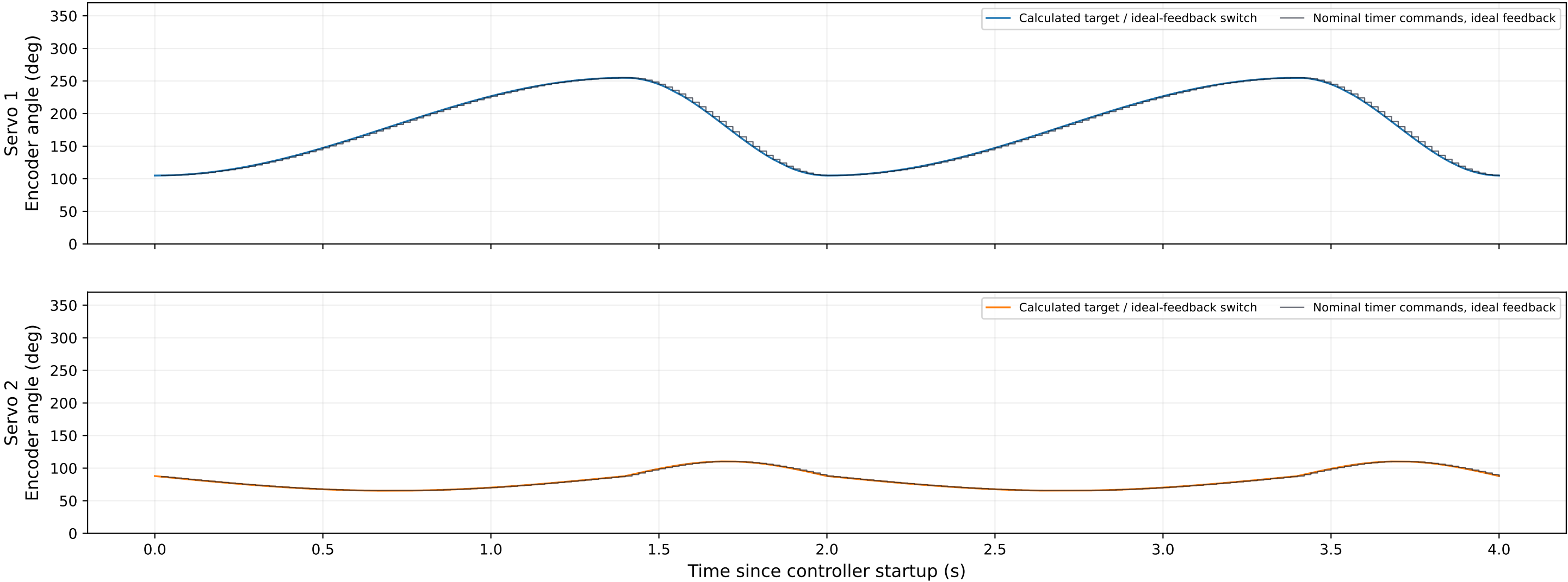
Run	f (Hz)	A1 (deg)	C1 (deg)	A2 / power (deg)	C2 (deg)	Direction	Power ratio	Extra
1	0.5	75.00	180.00	22.50	87.89	—	0.7	
2	0.5	75.00	180.00	30.00	87.89	—	0.7	
3	0.5	75.00	180.00	45.00	87.89	—	0.7	
4	0.5	75.00	180.00	45.00	87.89	—	0.5	
5	0.5	75.00	180.00	22.50	87.89	—	0.7	
6	0.5	75.00	180.00	30.00	87.89	—	0.7	
7	0.5	75.00	180.00	45.00	87.89	—	0.7	
8	0.5	75.00	180.00	45.00	87.89	—	0.5	

All angles above are degrees. Centers use encoder zero; amplitudes are offsets. Harmonic amplitudes are scale factors.
Degrees = counts × 360/4096. Raw count parameters are retained in the CSV/JSON and exact position equations.

Sin (LNU) | Sin (LU) / Run 1

f=0.5 Hz; S1 amplitude=75.00°; centers=180.00° / 87.89°; power=0.7

MODEL ONLY • Table settings + controller equations • No measured force or motion

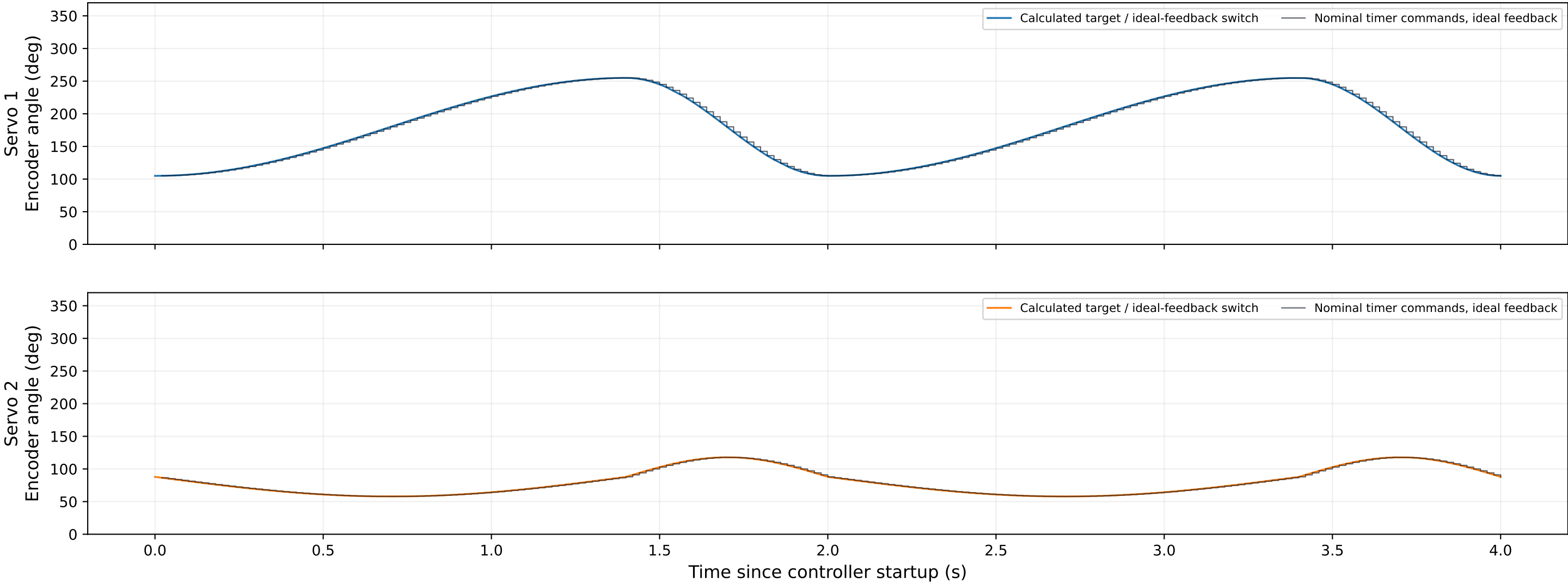


$u = (t \times 0.5) \bmod 1$; $b = 0.7$; $s = u/b$ during power, $(u-b)/(1-b)$ during recovery.
 $\theta_1 = -75 \cos(\pi s)$ during power; $+75 \cos(\pi s)$ during recovery.
 $\theta_2 = -22.5 \sin(\pi s)$ during power; $+22.5 \sin(\pi s)$ during recovery.
 $p_1 = \text{clip}(\text{int}(2048 + 4096\theta_1/360))$; $p_2 = \text{clip}(\text{int}(1000 + 4096\theta_2/360))$; clip to $[0, 4095]$.
Plotted encoder angle = $p \times 360/4096$ degrees. Above position equations retain raw counts for exact rounding.
Rows 5-8 repeat settings 1-4; retained as separate runs.

Sin (LNU) | Sin (LU) / Run 2

f=0.5 Hz; S1 amplitude=75.00°; centers=180.00° / 87.89°; power=0.7

MODEL ONLY • Table settings + controller equations • No measured force or motion

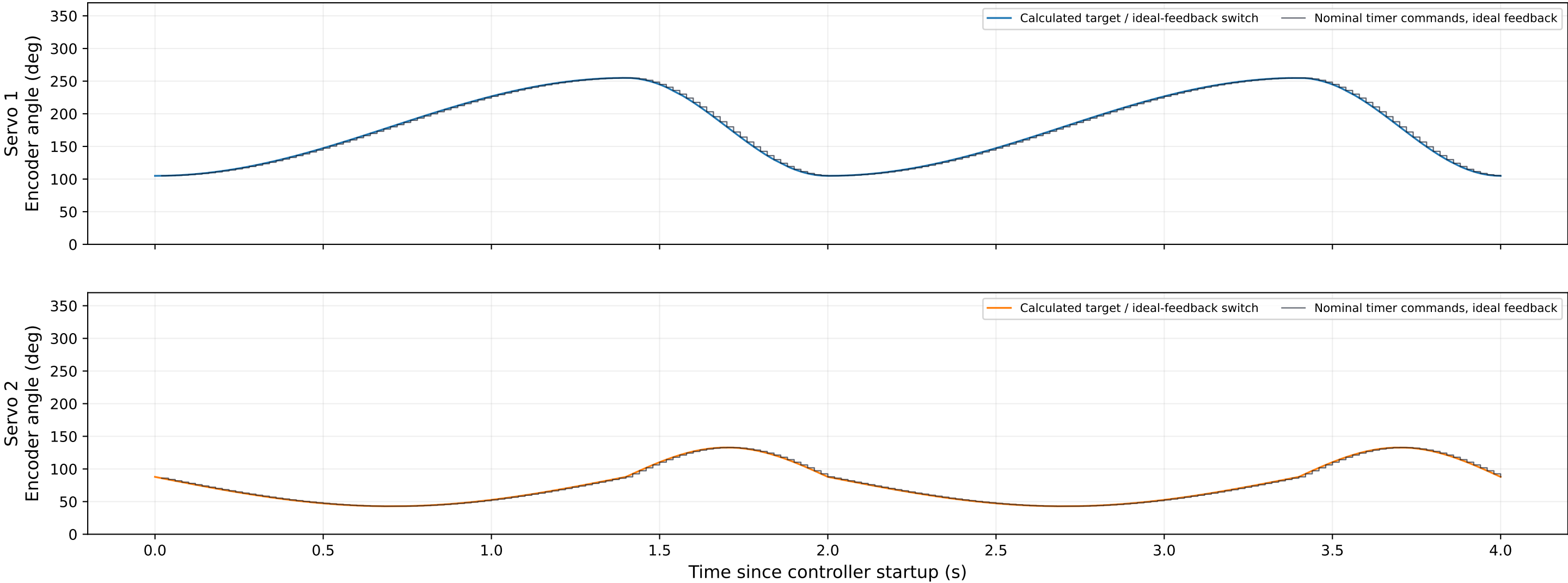


$u = (t \times 0.5) \bmod 1$; $b = 0.7$; $s = u/b$ during power, $(u-b)/(1-b)$ during recovery.
 $\theta_1 = -75 \cos(\pi s)$ during power; $+75 \cos(\pi s)$ during recovery.
 $\theta_2 = -30 \sin(\pi s)$ during power; $+30 \sin(\pi s)$ during recovery.
 $p_1 = \text{clip}(\text{int}(2048 + 4096\theta_1/360))$; $p_2 = \text{clip}(\text{int}(1000 + 4096\theta_2/360))$; clip to $[0, 4095]$.
Plotted encoder angle = $p \times 360/4096$ degrees. Above position equations retain raw counts for exact rounding.
Rows 5-8 repeat settings 1-4; retained as separate runs.

Sin (LNU) | Sin (LU) / Run 3

f=0.5 Hz; S1 amplitude=75.00°; centers=180.00° / 87.89°; power=0.7

MODEL ONLY • Table settings + controller equations • No measured force or motion

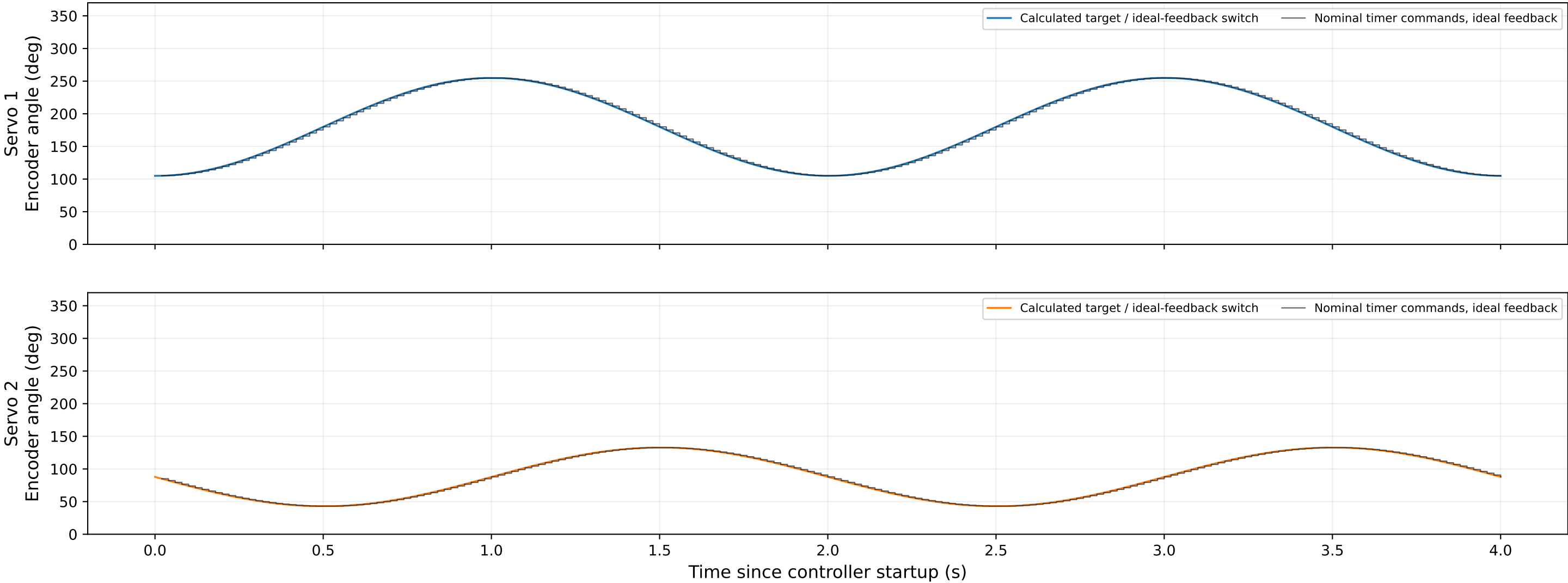


$u = (t \times 0.5) \bmod 1$; $b = 0.7$; $s = u/b$ during power, $(u-b)/(1-b)$ during recovery.
 $\theta_1 = -75 \cos(\pi s)$ during power; $+75 \cos(\pi s)$ during recovery.
 $\theta_2 = -45 \sin(\pi s)$ during power; $+45 \sin(\pi s)$ during recovery.
 $p_1 = \text{clip}(\text{int}(2048 + 4096\theta_1/360))$; $p_2 = \text{clip}(\text{int}(1000 + 4096\theta_2/360))$; clip to $[0, 4095]$.
Plotted encoder angle = $p \times 360/4096$ degrees. Above position equations retain raw counts for exact rounding.
Rows 5-8 repeat settings 1-4; retained as separate runs.

Sin (LNU) | Sin (LU) / Run 4

f=0.5 Hz; S1 amplitude=75.00°; centers=180.00° / 87.89°; power=0.5

MODEL ONLY • Table settings + controller equations • No measured force or motion

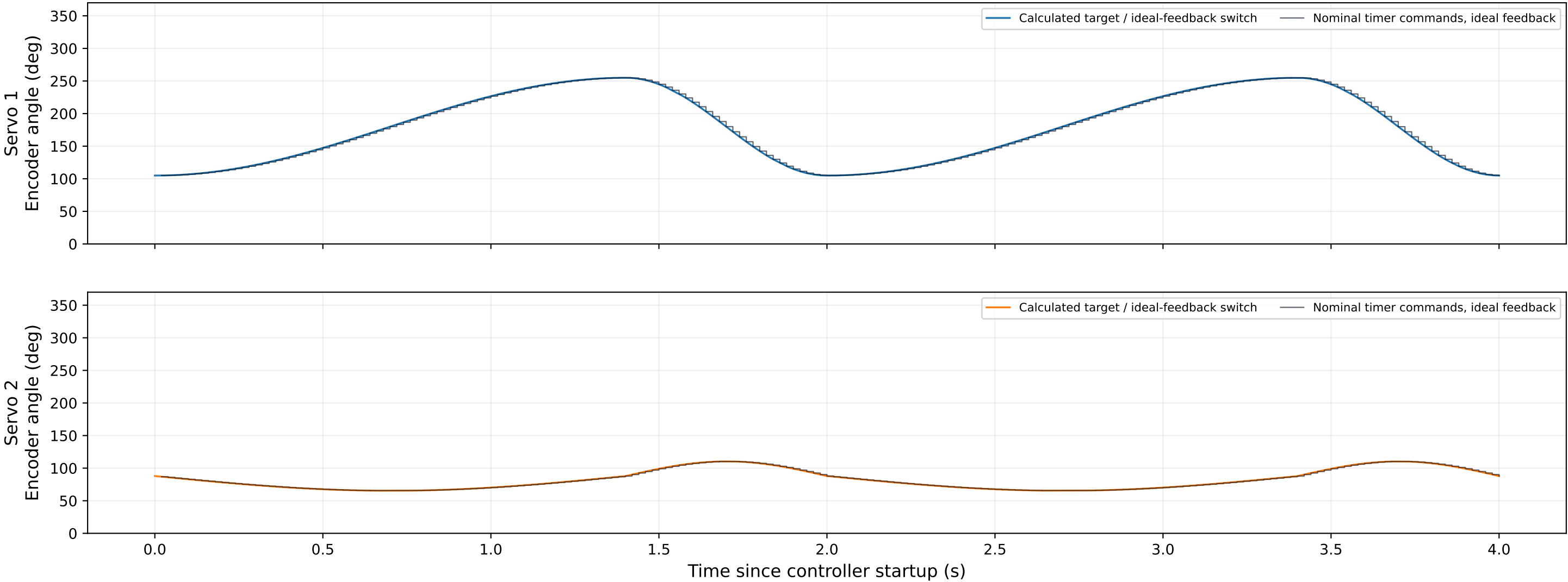


$u = (t \times 0.5) \bmod 1$; $b = 0.5$; $s = u/b$ during power, $(u-b)/(1-b)$ during recovery.
 $\theta_1 = -75 \cos(\pi s)$ during power; $+75 \cos(\pi s)$ during recovery.
 $\theta_2 = -45 \sin(\pi s)$ during power; $+45 \sin(\pi s)$ during recovery.
 $p_1 = \text{clip}(\text{int}(2048 + 4096\theta_1/360))$; $p_2 = \text{clip}(\text{int}(1000 + 4096\theta_2/360))$; clip to $[0, 4095]$.
Plotted encoder angle = $p \times 360/4096$ degrees. Above position equations retain raw counts for exact rounding.
Rows 5-8 repeat settings 1-4; retained as separate runs.

Sin (LNU) | Sin (LU) / Run 5

f=0.5 Hz; S1 amplitude=75.00°; centers=180.00° / 87.89°; power=0.7

MODEL ONLY • Table settings + controller equations • No measured force or motion

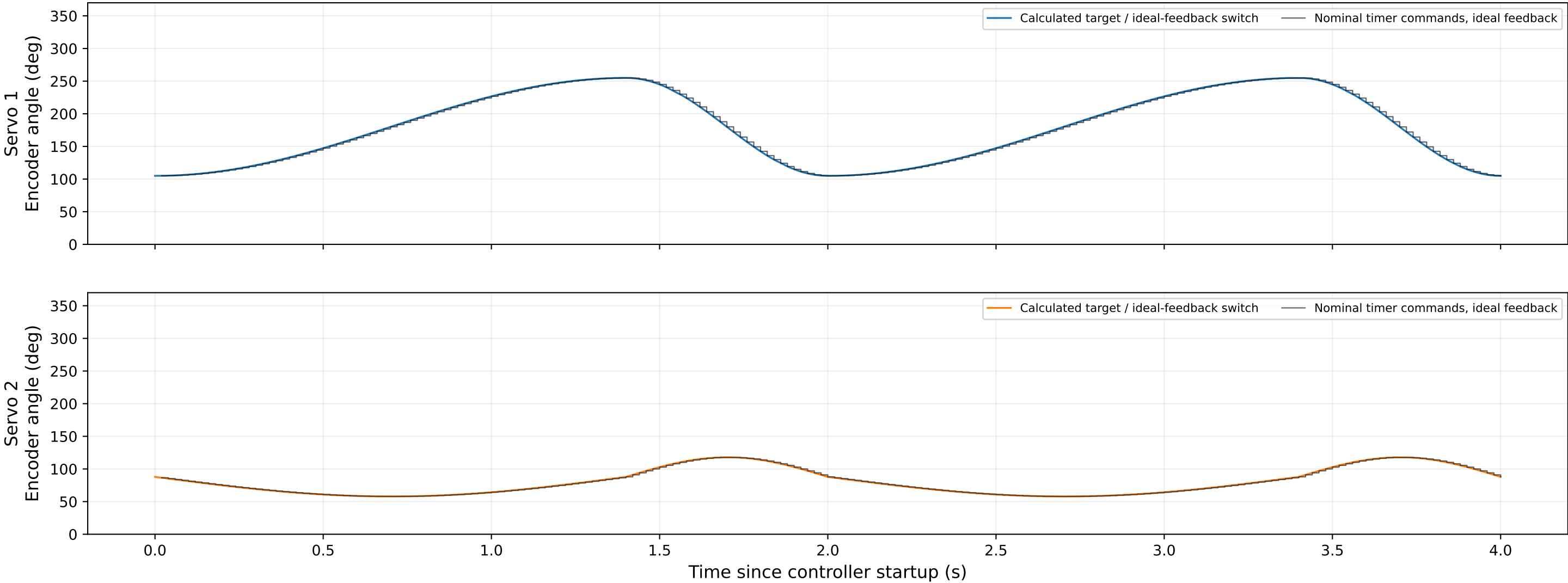


$u = (t \times 0.5) \bmod 1$; $b = 0.7$; $s = u/b$ during power, $(u-b)/(1-b)$ during recovery.
 $\theta_1 = -75 \cos(\pi s)$ during power; $+75 \cos(\pi s)$ during recovery.
 $\theta_2 = -22.5 \sin(\pi s)$ during power; $+22.5 \sin(\pi s)$ during recovery.
 $p_1 = \text{clip}(\text{int}(2048 + 4096\theta_1/360))$; $p_2 = \text{clip}(\text{int}(1000 + 4096\theta_2/360))$; clip to $[0, 4095]$.
Plotted encoder angle = $p \times 360/4096$ degrees. Above position equations retain raw counts for exact rounding.
Rows 5-8 repeat settings 1-4; retained as separate runs.

Sin (LNU) | Sin (LU) / Run 6

f=0.5 Hz; S1 amplitude=75.00°; centers=180.00° / 87.89°; power=0.7

MODEL ONLY • Table settings + controller equations • No measured force or motion

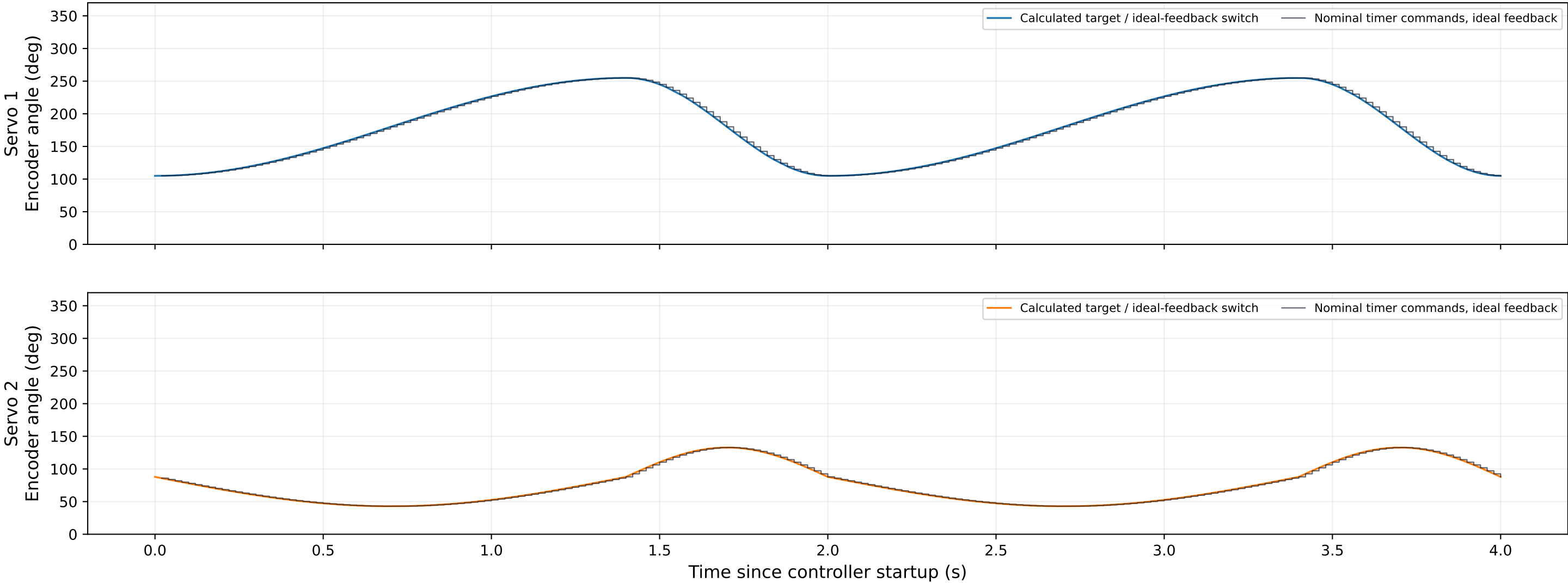


$u = (t \times 0.5) \bmod 1$; $b = 0.7$; $s = u/b$ during power, $(u-b)/(1-b)$ during recovery.
 $\theta_1 = -75 \cos(\pi s)$ during power; $+75 \cos(\pi s)$ during recovery.
 $\theta_2 = -30 \sin(\pi s)$ during power; $+30 \sin(\pi s)$ during recovery.
 $p_1 = \text{clip}(\text{int}(2048 + 4096\theta_1/360))$; $p_2 = \text{clip}(\text{int}(1000 + 4096\theta_2/360))$; clip to $[0, 4095]$.
Plotted encoder angle = $p \times 360/4096$ degrees. Above position equations retain raw counts for exact rounding.
Rows 5-8 repeat settings 1-4; retained as separate runs.

Sin (LNU) | Sin (LU) / Run 7

f=0.5 Hz; S1 amplitude=75.00°; centers=180.00° / 87.89°; power=0.7

MODEL ONLY • Table settings + controller equations • No measured force or motion

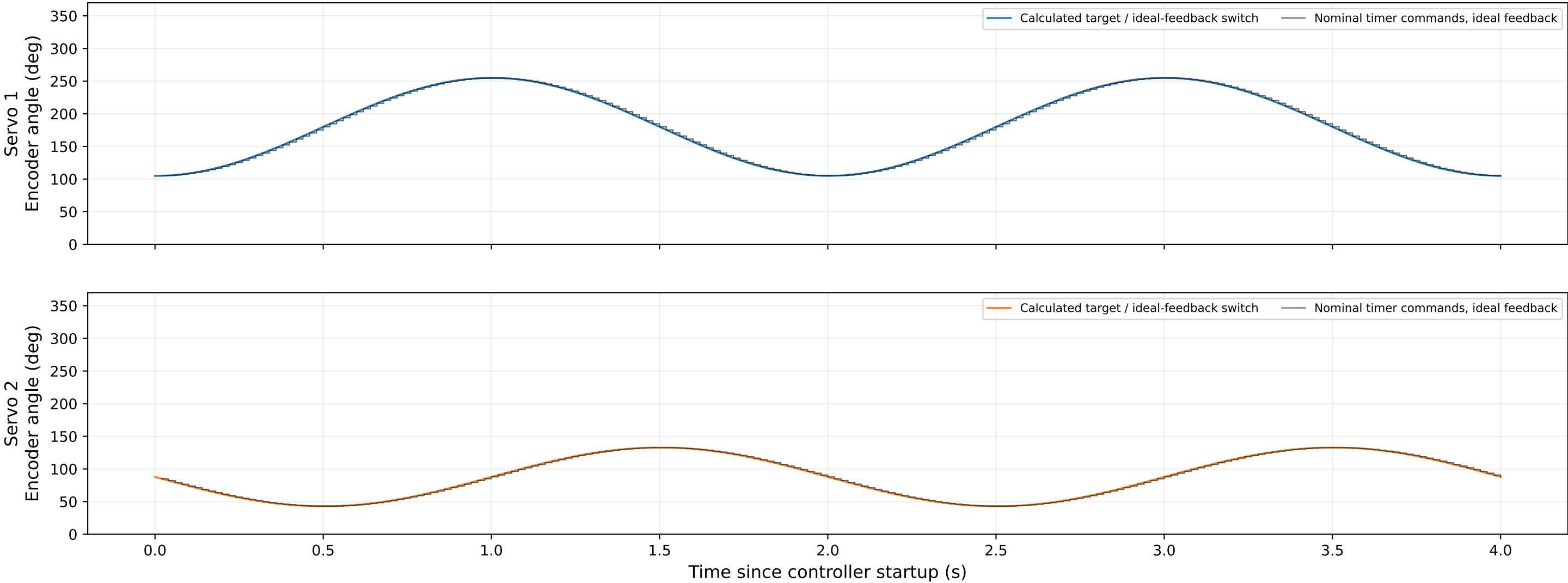


$u = (t \times 0.5) \bmod 1$; $b = 0.7$; $s = u/b$ during power, $(u-b)/(1-b)$ during recovery.
 $\theta_1 = -75 \cos(\pi s)$ during power; $+75 \cos(\pi s)$ during recovery.
 $\theta_2 = -45 \sin(\pi s)$ during power; $+45 \sin(\pi s)$ during recovery.
 $p_1 = \text{clip}(\text{int}(2048 + 4096\theta_1/360))$; $p_2 = \text{clip}(\text{int}(1000 + 4096\theta_2/360))$; clip to $[0, 4095]$.
Plotted encoder angle = $p \times 360/4096$ degrees. Above position equations retain raw counts for exact rounding.
Rows 5-8 repeat settings 1-4; retained as separate runs.

Sin (LNU) | Sin (LU) / Run 8

f=0.5 Hz; S1 amplitude=75.00°; centers=180.00° / 87.89°; power=0.5

MODEL ONLY • Table settings + controller equations • No measured force or motion



$u = (t \times 0.5) \bmod 1$; $b = 0.5$; $s = u/b$ during power, $(u-b)/(1-b)$ during recovery.
 $\theta_1 = -75 \cos(\pi s)$ during power; $+75 \cos(\pi s)$ during recovery.
 $\theta_2 = -45 \sin(\pi s)$ during power; $+45 \sin(\pi s)$ during recovery.
 $p_1 = \text{clip}(\text{int}(2048 + 4096\theta_1/360))$; $p_2 = \text{clip}(\text{int}(1000 + 4096\theta_2/360))$; clip to [0,4095].
Plotted encoder angle = $p \times 360/4096$ degrees. Above position equations retain raw counts for exact rounding.
Rows 5-8 repeat settings 1-4; retained as separate runs.

Sin (U) | Square (R)

Original key: SIN / PDF p.6 / rows map to run numbers

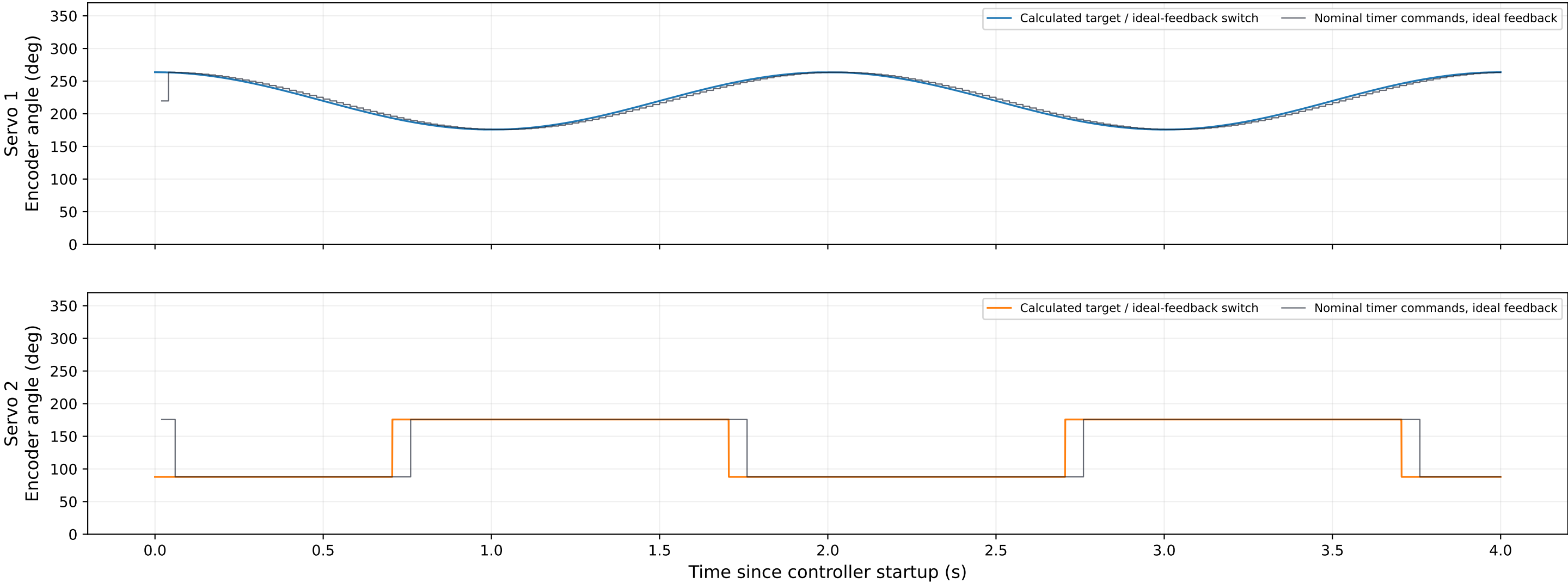
Run	f (Hz)	A1 (deg)	C1 (deg)	A2 / power (deg)	C2 (deg)	Direction	Power ratio	Extra
1	0.5	43.95	219.73	87.9/175.8	175.78	1	—	limits 193.4/246.1
2	0.75	43.95	219.73	87.9/175.8	175.78	1	—	limits 193.4/246.1

All angles above are degrees. Centers use encoder zero; amplitudes are offsets. Harmonic amplitudes are scale factors.
Degrees = counts × 360/4096. Raw count parameters are retained in the CSV/JSON and exact position equations.

Sin (U) | Square (R) / Run 1

f=0.5 Hz; S1 amplitude=43.95°; centers=219.73° / 175.78°; direction=1

MODEL ONLY • Table settings + controller equations • No measured force or motion

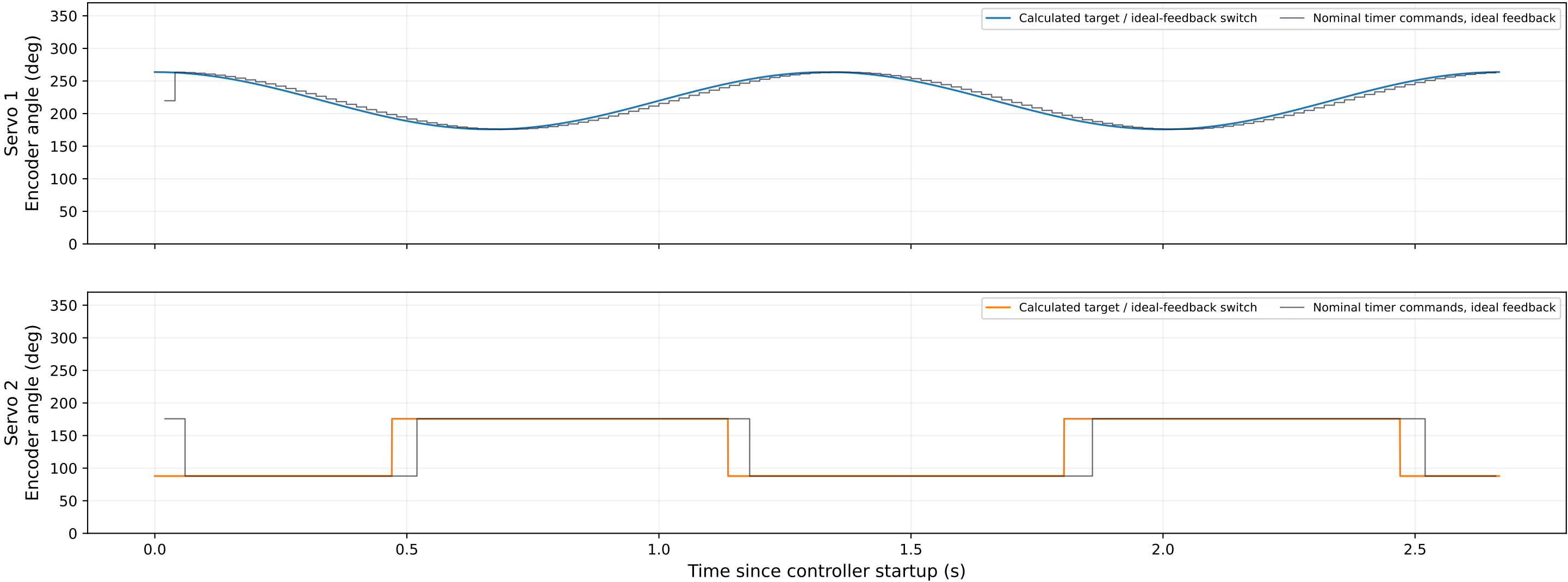


$p1 = \text{round}(2500 + 500 \cos(1\pi t))$
 $q2[k+1] = 1000$ if $e1[k] > 2800$; 2000 if $e1[k] < 2200$; otherwise $q2[k]$.
Plotted encoder angle = $p \times 360/4096$ degrees. Above position equations retain raw counts for exact rounding.

Sin (U) | Square (R) / Run 2

f=0.75 Hz; S1 amplitude=43.95°; centers=219.73° / 175.78°; direction=1

MODEL ONLY • Table settings + controller equations • No measured force or motion



$p1 = \text{round}(2500 + 500 \cos(1.5\pi t))$
 $q2[k+1] = 1000$ if $e1[k] > 2800$; 2000 if $e1[k] < 2200$; otherwise $q2[k]$.
Plotted encoder angle = $p \times 360/4096$ degrees. Above position equations retain raw counts for exact rounding.

Sin (HU) | Square (R)

Original key: SINmiddle / PDF p.8 / rows map to run numbers

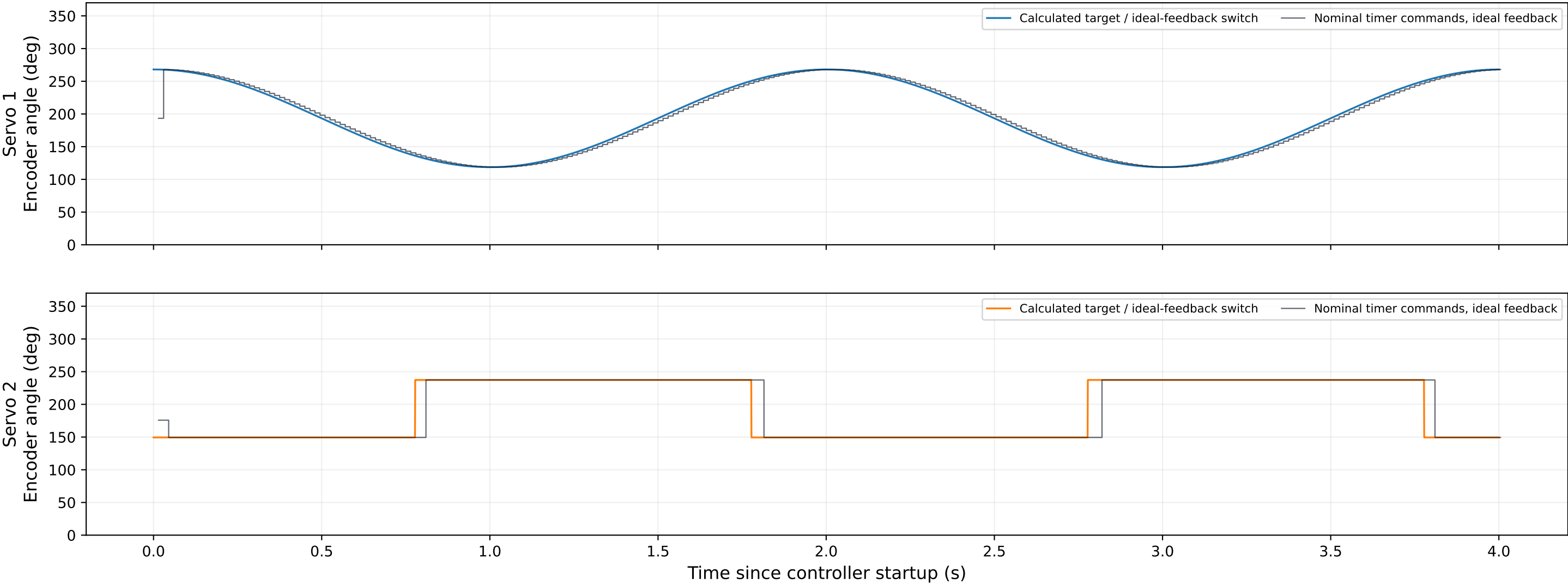
Run	f (Hz)	A1 (deg)	C1 (deg)	A2 / power (deg)	C2 (deg)	Direction	Power ratio	Extra
1	0.5	74.71	193.36	149.4/237.3	175.78	1	—	limits 136.2/250.5

All angles above are degrees. Centers use encoder zero; amplitudes are offsets. Harmonic amplitudes are scale factors.
Degrees = counts × 360/4096. Raw count parameters are retained in the CSV/JSON and exact position equations.

Sin (HU) | Square (R) / Run 1

f=0.5 Hz; S1 amplitude=74.71°; centers=193.36° / 175.78°; direction=1

MODEL ONLY • Table settings + controller equations • No measured force or motion



$p1 = \text{round}(2200 + 850 \cos(1\pi t))$
 $q2[k+1] = 1700$ if $e1[k] > 2850$; 2700 if $e1[k] < 1550$; otherwise $q2[k]$.
Plotted encoder angle = $p \times 360/4096$ degrees. Above position equations retain raw counts for exact rounding.

Sin (LU) | Square (LNR)

Original key: yawPower / PDF p.4 / rows map to run numbers

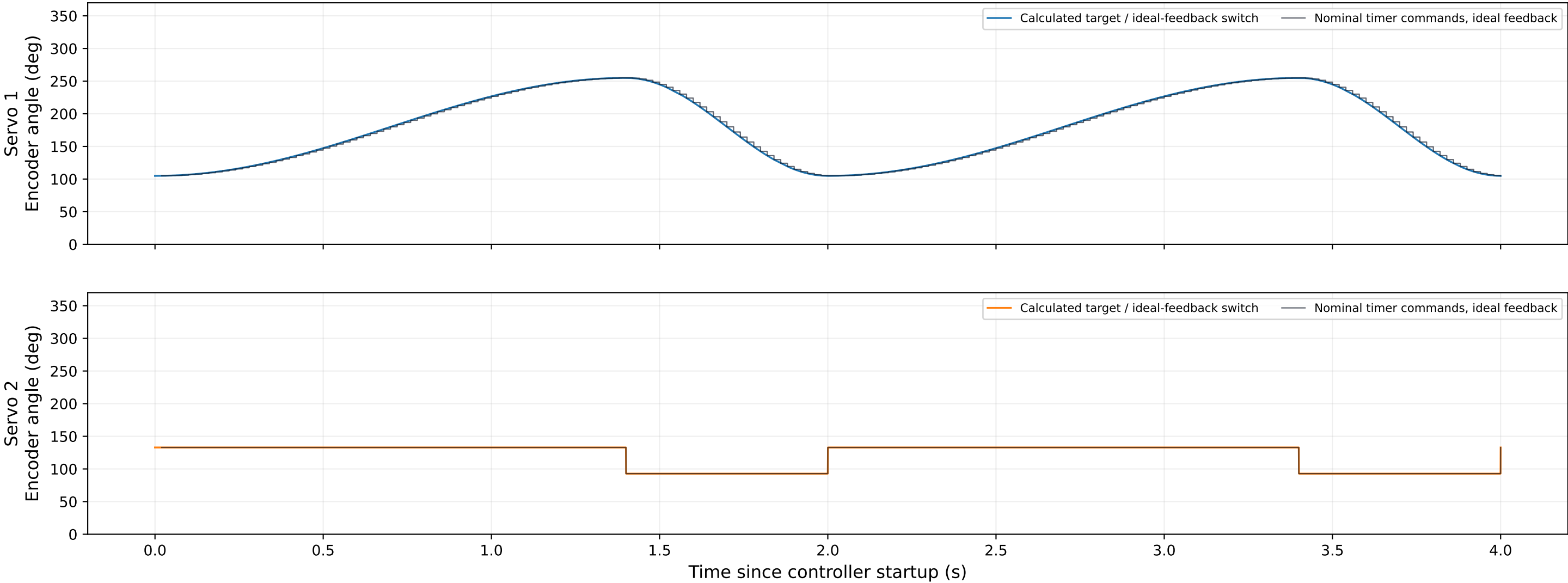
Run	f (Hz)	A1 (deg)	C1 (deg)	A2 / power (deg)	C2 (deg)	Direction	Power ratio	Extra
1	0.5	75.00	180.00	45.00	87.89	—	0.7	recovery 5 deg
2	0.5	75.00	180.00	30.00	87.89	—	0.7	recovery 5 deg
3	0.5	75.00	180.00	60.00	87.89	—	0.7	recovery 5 deg
4	0.5	75.00	180.00	90.00	87.89	—	0.7	recovery 5 deg

All angles above are degrees. Centers use encoder zero; amplitudes are offsets. Harmonic amplitudes are scale factors.
Degrees = counts × 360/4096. Raw count parameters are retained in the CSV/JSON and exact position equations.

Sin (LU) | Square (LNR) / Run 1

f=0.5 Hz; S1 amplitude=75.00°; centers=180.00° / 87.89°; power=0.7

MODEL ONLY • Table settings + controller equations • No measured force or motion

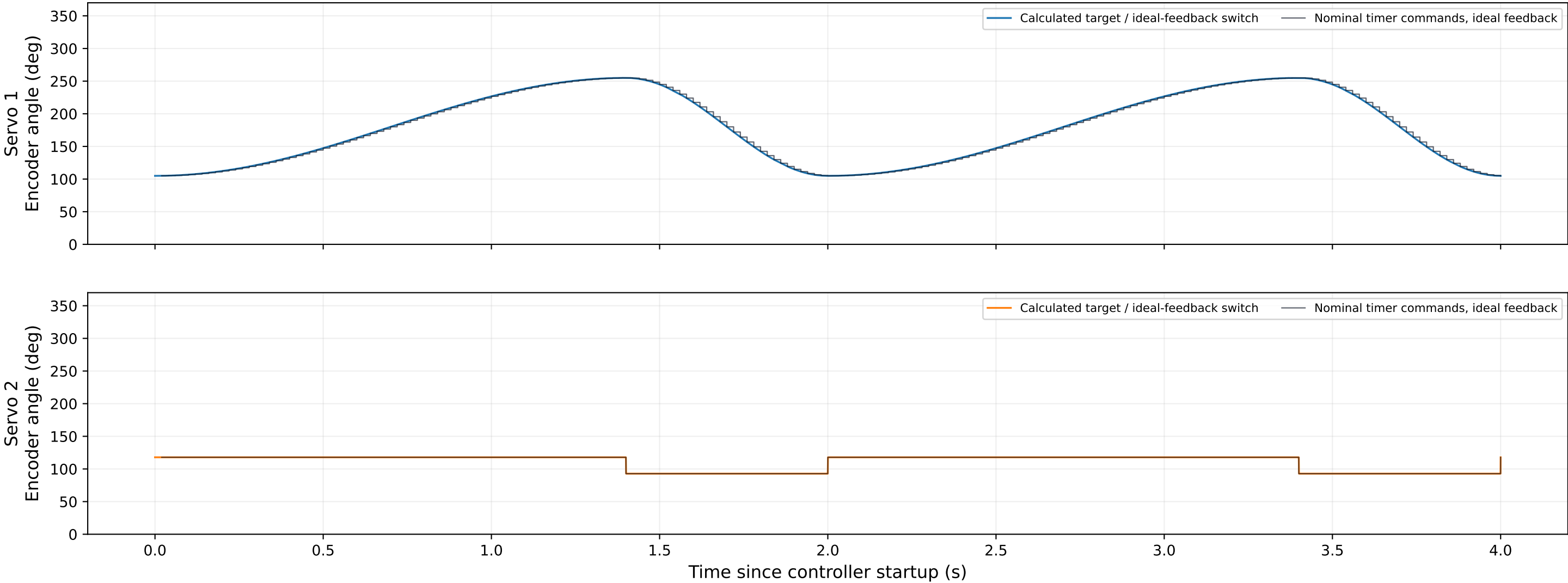


$u = (t \times 0.5) \bmod 1$; $b = 0.7$; $s = u/b$ during power, $(u-b)/(1-b)$ during recovery.
 $\theta_1 = -75 \cos(\pi s)$ during power; $+75 \cos(\pi s)$ during recovery.
 $\theta_2 = 45^\circ$ during power; 5° during recovery.
 $p_1 = \text{clip}(\text{int}(2048 + 4096\theta_1/360))$; $p_2 = \text{clip}(\text{int}(1000 + 4096\theta_2/360))$; clip to $[0, 4095]$.
Plotted encoder angle = $p \times 360/4096$ degrees. Above position equations retain raw counts for exact rounding.

Sin (LU) | Square (LNR) / Run 2

f=0.5 Hz; S1 amplitude=75.00°; centers=180.00° / 87.89°; power=0.7

MODEL ONLY • Table settings + controller equations • No measured force or motion

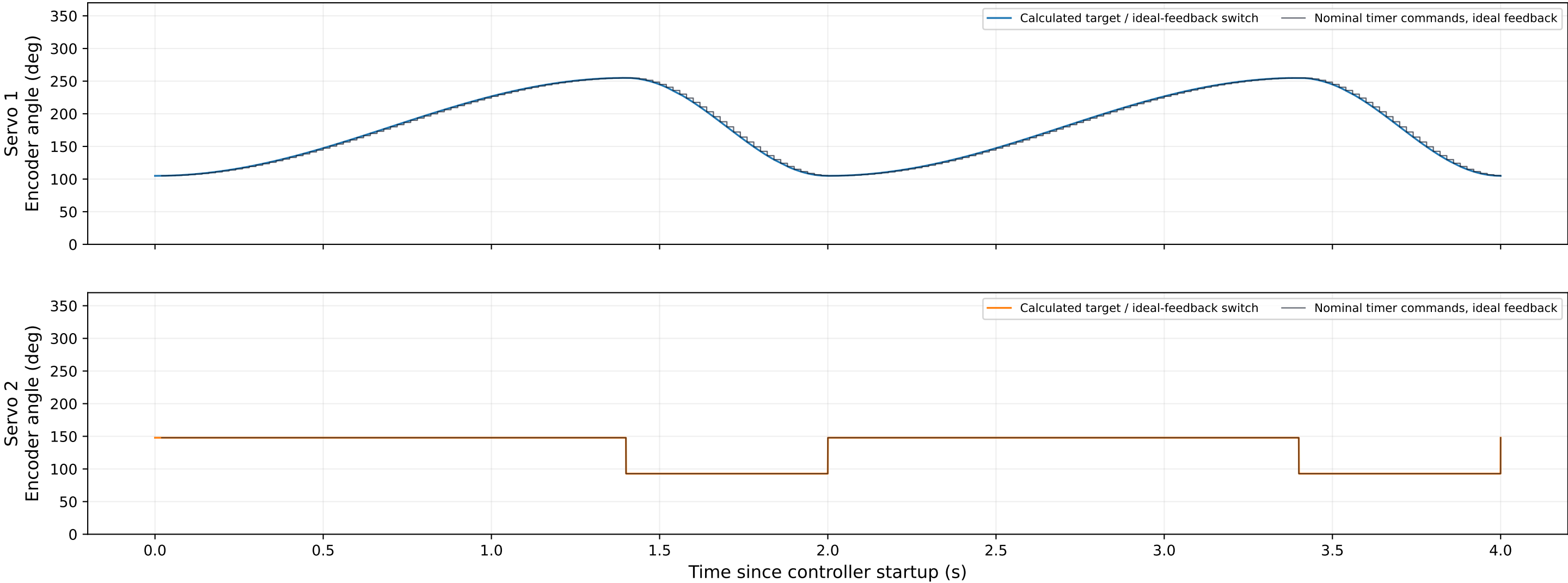


$u = (t \times 0.5) \bmod 1$; $b = 0.7$; $s = u/b$ during power, $(u-b)/(1-b)$ during recovery.
 $\theta_1 = -75 \cos(\pi s)$ during power; $+75 \cos(\pi s)$ during recovery.
 $\theta_2 = 30^\circ$ during power; 5° during recovery.
 $p_1 = \text{clip}(\text{int}(2048 + 4096\theta_1/360))$; $p_2 = \text{clip}(\text{int}(1000 + 4096\theta_2/360))$; clip to $[0, 4095]$.
Plotted encoder angle = $p \times 360/4096$ degrees. Above position equations retain raw counts for exact rounding.

Sin (LU) | Square (LNR) / Run 3

f=0.5 Hz; S1 amplitude=75.00°; centers=180.00° / 87.89°; power=0.7

MODEL ONLY • Table settings + controller equations • No measured force or motion

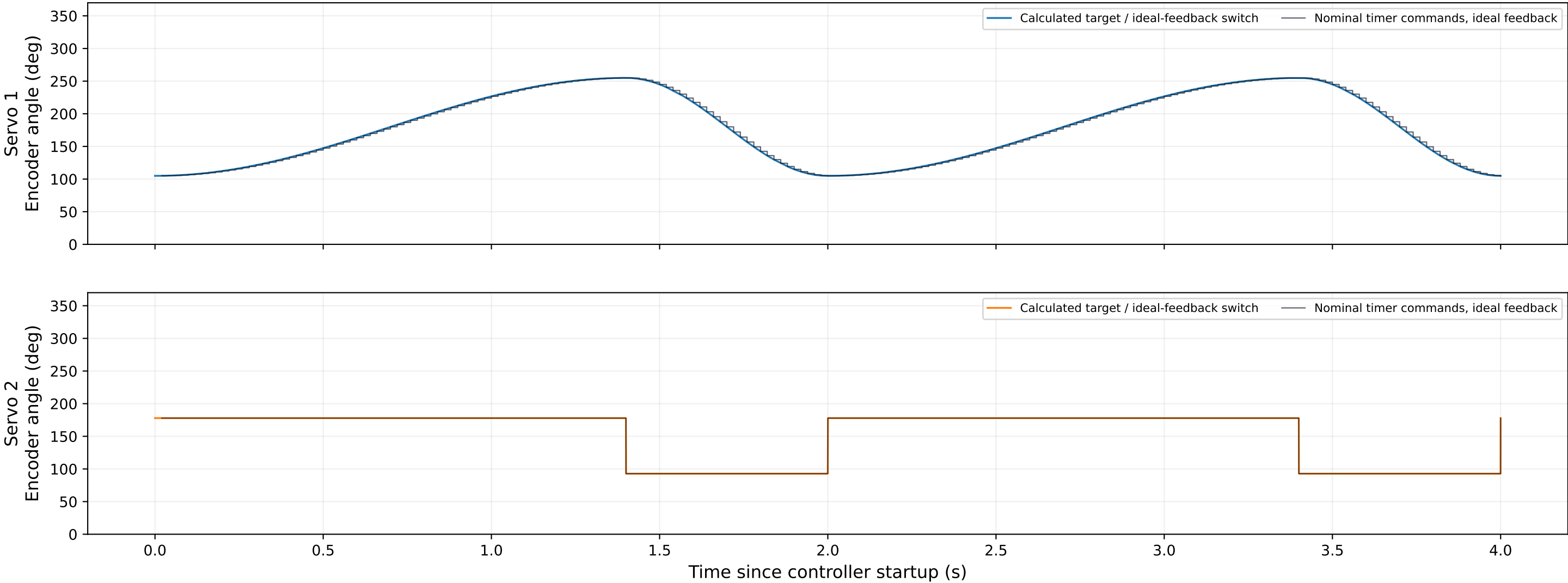


$u = (t \times 0.5) \bmod 1$; $b = 0.7$; $s = u/b$ during power, $(u-b)/(1-b)$ during recovery.
 $\theta_1 = -75 \cos(\pi s)$ during power; $+75 \cos(\pi s)$ during recovery.
 $\theta_2 = 60^\circ$ during power; 5° during recovery.
 $p_1 = \text{clip}(\text{int}(2048 + 4096\theta_1/360))$; $p_2 = \text{clip}(\text{int}(1000 + 4096\theta_2/360))$; clip to $[0, 4095]$.
Plotted encoder angle = $p \times 360/4096$ degrees. Above position equations retain raw counts for exact rounding.

Sin (LU) | Square (LNR) / Run 4

f=0.5 Hz; S1 amplitude=75.00°; centers=180.00° / 87.89°; power=0.7

MODEL ONLY • Table settings + controller equations • No measured force or motion



$u = (t \times 0.5) \bmod 1$; $b = 0.7$; $s = u/b$ during power, $(u-b)/(1-b)$ during recovery.
 $\theta_1 = -75 \cos(\pi s)$ during power; $+75 \cos(\pi s)$ during recovery.
 $\theta_2 = 90^\circ$ during power; 5° during recovery.
 $p_1 = \text{clip}(\text{int}(2048 + 4096\theta_1/360))$; $p_2 = \text{clip}(\text{int}(1000 + 4096\theta_2/360))$; clip to $[0, 4095]$.
Plotted encoder angle = $p \times 360/4096$ degrees. Above position equations retain raw counts for exact rounding.

Sin (LLU) | Square (LNR)

Original key: rollAmpAndFrequency / PDF p.3 / rows map to run numbers

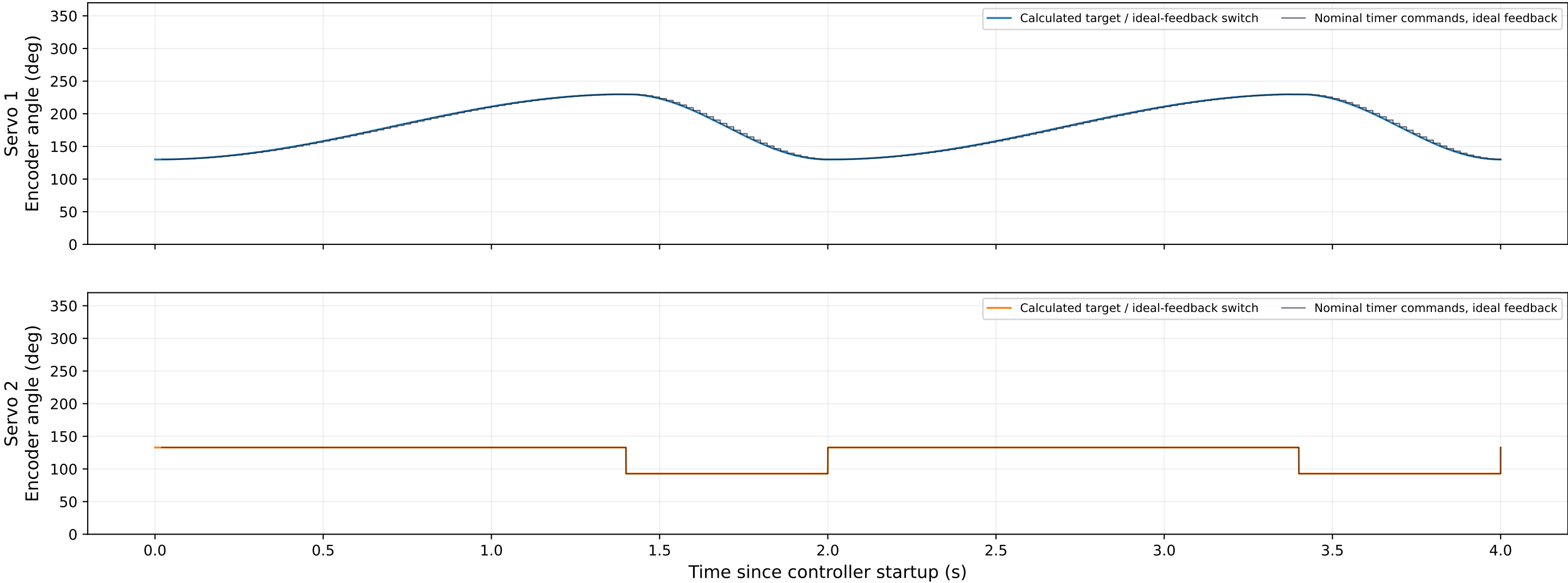
Run	f (Hz)	A1 (deg)	C1 (deg)	A2 / power (deg)	C2 (deg)	Direction	Power ratio	Extra
1	0.5	50.00	180.00	45.00	87.89	—	0.7	recovery 5 deg
2	0.75	50.00	180.00	45.00	87.89	—	0.7	recovery 5 deg
3	1	50.00	180.00	45.00	87.89	—	0.7	recovery 5 deg
4	1	60.00	180.00	45.00	87.89	—	0.7	recovery 5 deg
5	0.75	60.00	180.00	45.00	87.89	—	0.7	recovery 5 deg
6	0.5	60.00	180.00	45.00	87.89	—	0.7	recovery 5 deg
7	0.5	70.00	180.00	45.00	87.89	—	0.7	recovery 5 deg
8	0.75	70.00	180.00	45.00	87.89	—	0.7	recovery 5 deg
9	1	70.00	180.00	45.00	87.89	—	0.7	recovery 5 deg
10	0.5	80.00	180.00	45.00	87.89	—	0.7	recovery 5 deg
11	0.75	80.00	180.00	45.00	87.89	—	0.7	recovery 5 deg
12	1	80.00	180.00	45.00	87.89	—	0.7	recovery 5 deg

All angles above are degrees. Centers use encoder zero; amplitudes are offsets. Harmonic amplitudes are scale factors.
Degrees = counts × 360/4096. Raw count parameters are retained in the CSV/JSON and exact position equations.

Sin (LLU) | Square (LNR) / Run 1

f=0.5 Hz; S1 amplitude=50.00°; centers=180.00° / 87.89°; power=0.7

MODEL ONLY • Table settings + controller equations • No measured force or motion

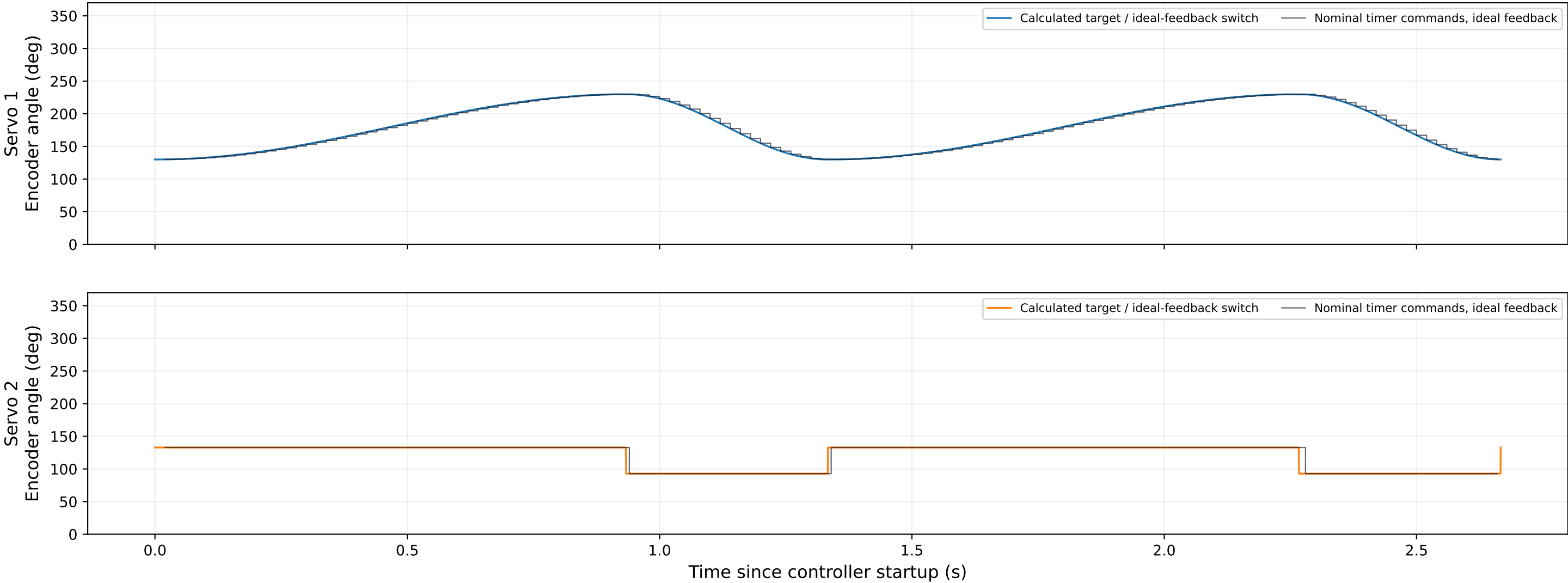


$u = (t \times 0.5) \bmod 1$; $b = 0.7$; $s = u/b$ during power, $(u-b)/(1-b)$ during recovery.
 $\theta_1 = -50 \cos(\pi s)$ during power; $+50 \cos(\pi s)$ during recovery.
 $\theta_2 = 45^\circ$ during power; 5° during recovery.
 $p_1 = \text{clip}(\text{int}(2048 + 4096\theta_1/360))$; $p_2 = \text{clip}(\text{int}(1000 + 4096\theta_2/360))$; clip to $[0, 4095]$.
Plotted encoder angle = $p \times 360/4096$ degrees. Above position equations retain raw counts for exact rounding.

Sin (LLU) | Square (LNR) / Run 2

f=0.75 Hz; S1 amplitude=50.00°; centers=180.00° / 87.89°; power=0.7

MODEL ONLY • Table settings + controller equations • No measured force or motion

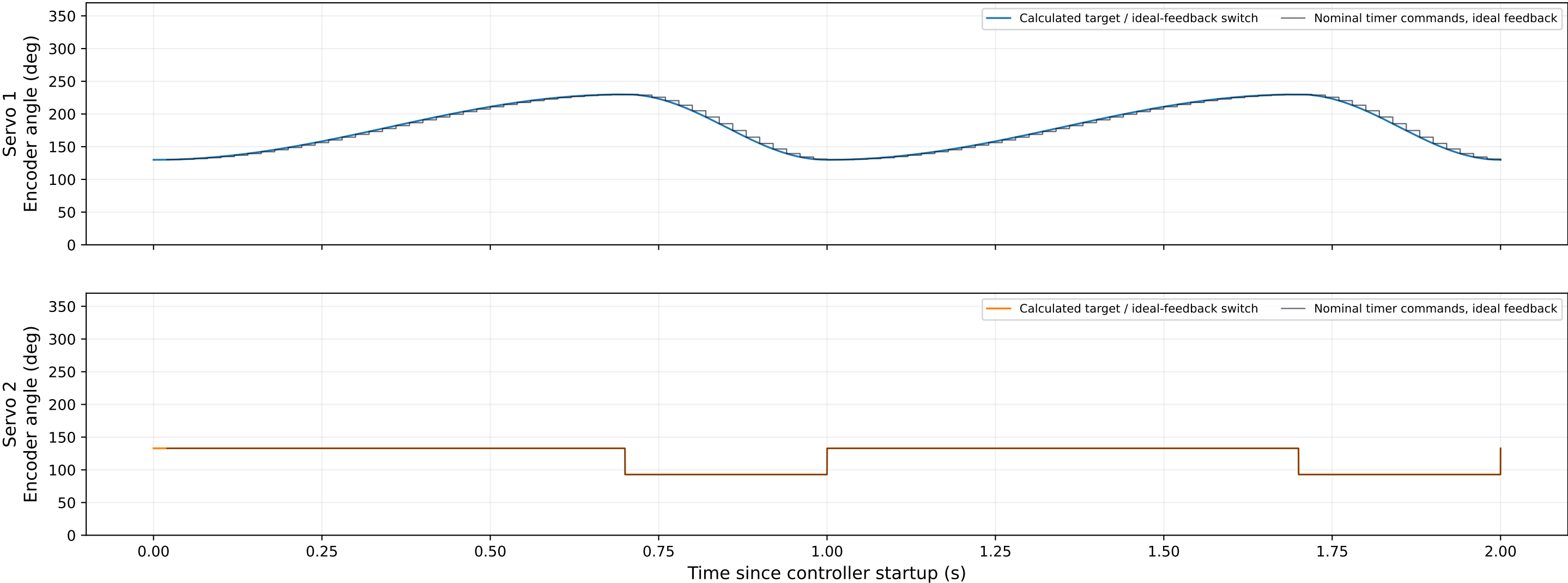


$u = (t \times 0.75) \bmod 1$; $b = 0.7$; $s = u/b$ during power, $(u-b)/(1-b)$ during recovery.
 $\theta_1 = -50 \cos(\pi s)$ during power; $+50 \cos(\pi s)$ during recovery.
 $\theta_2 = 45^\circ$ during power; 5° during recovery.
 $p_1 = \text{clip}(\text{int}(2048 + 4096\theta_1/360))$; $p_2 = \text{clip}(\text{int}(1000 + 4096\theta_2/360))$; clip to $[0, 4095]$.
Plotted encoder angle = $p \times 360/4096$ degrees. Above position equations retain raw counts for exact rounding.

Sin (LLU) | Square (LNR) / Run 3

f=1 Hz; S1 amplitude=50.00°; centers=180.00° / 87.89°; power=0.7

MODEL ONLY • Table settings + controller equations • No measured force or motion

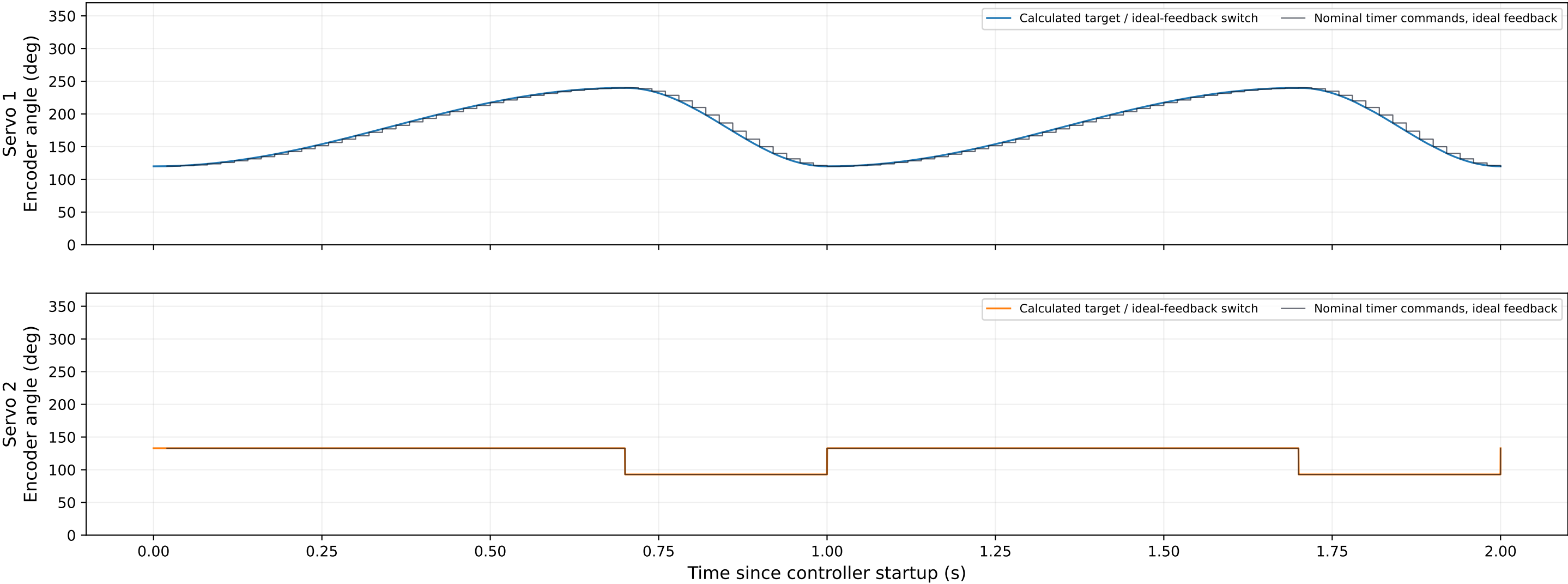


$u = (t \times 1) \bmod 1$; $b = 0.7$; $s = u/b$ during power, $(u-b)/(1-b)$ during recovery.
 $\theta_1 = -50 \cos(\pi s)$ during power; $+50 \cos(\pi s)$ during recovery.
 $\theta_2 = 45^\circ$ during power; 5° during recovery.
 $p_1 = \text{clip}(\text{int}(2048 + 4096\theta_1/360))$; $p_2 = \text{clip}(\text{int}(1000 + 4096\theta_2/360))$; clip to $[0, 4095]$.
Plotted encoder angle = $p \times 360/4096$ degrees. Above position equations retain raw counts for exact rounding.

Sin (LLU) | Square (LNR) / Run 4

f=1 Hz; S1 amplitude=60.00°; centers=180.00° / 87.89°; power=0.7

MODEL ONLY • Table settings + controller equations • No measured force or motion

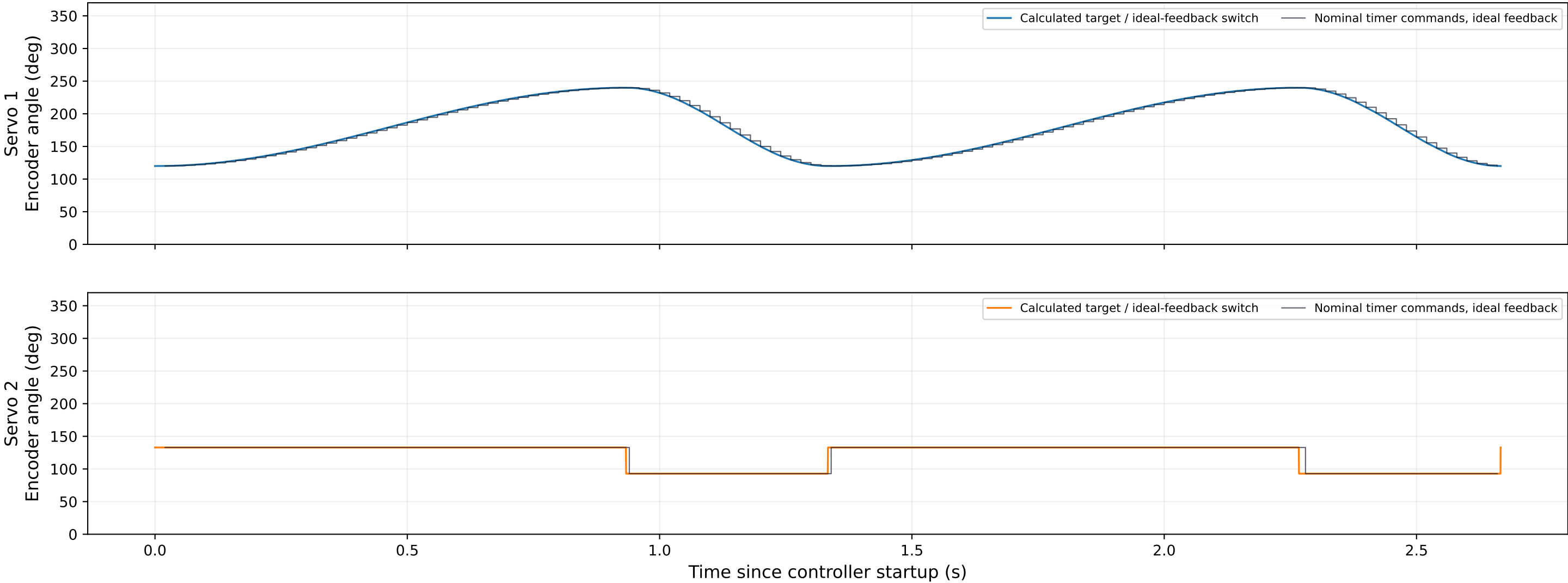


$u = (t \times 1) \bmod 1$; $b = 0.7$; $s = u/b$ during power, $(u-b)/(1-b)$ during recovery.
 $\theta_1 = -60 \cos(\pi s)$ during power; $+60 \cos(\pi s)$ during recovery.
 $\theta_2 = 45^\circ$ during power; 5° during recovery.
 $p_1 = \text{clip}(\text{int}(2048 + 4096\theta_1/360))$; $p_2 = \text{clip}(\text{int}(1000 + 4096\theta_2/360))$; clip to $[0, 4095]$.
Plotted encoder angle = $p \times 360/4096$ degrees. Above position equations retain raw counts for exact rounding.

Sin (LLU) | Square (LNR) / Run 5

f=0.75 Hz; S1 amplitude=60.00°; centers=180.00° / 87.89°; power=0.7

MODEL ONLY • Table settings + controller equations • No measured force or motion

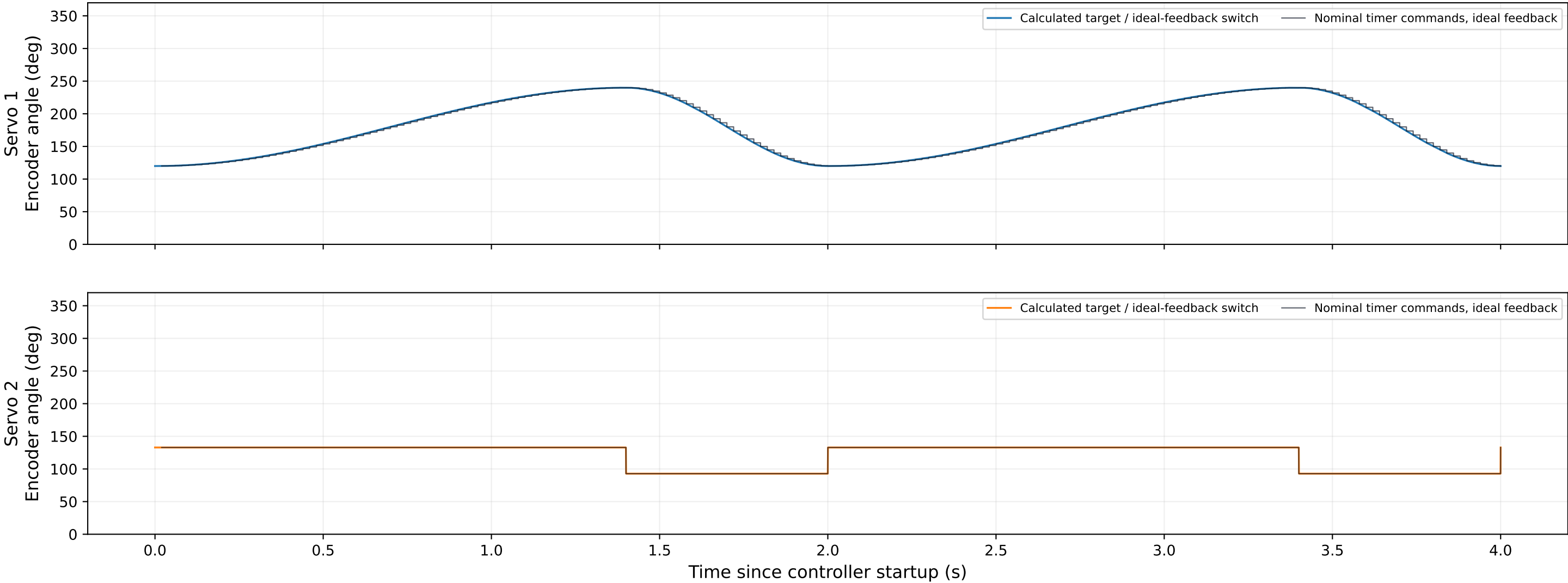


$u = (t \times 0.75) \bmod 1$; $b = 0.7$; $s = u/b$ during power, $(u-b)/(1-b)$ during recovery.
 $\theta_1 = -60 \cos(\pi s)$ during power; $+60 \cos(\pi s)$ during recovery.
 $\theta_2 = 45^\circ$ during power; 5° during recovery.
 $p_1 = \text{clip}(\text{int}(2048 + 4096\theta_1/360))$; $p_2 = \text{clip}(\text{int}(1000 + 4096\theta_2/360))$; clip to $[0, 4095]$.
Plotted encoder angle = $p \times 360/4096$ degrees. Above position equations retain raw counts for exact rounding.

Sin (LLU) | Square (LNR) / Run 6

f=0.5 Hz; S1 amplitude=60.00°; centers=180.00° / 87.89°; power=0.7

MODEL ONLY • Table settings + controller equations • No measured force or motion

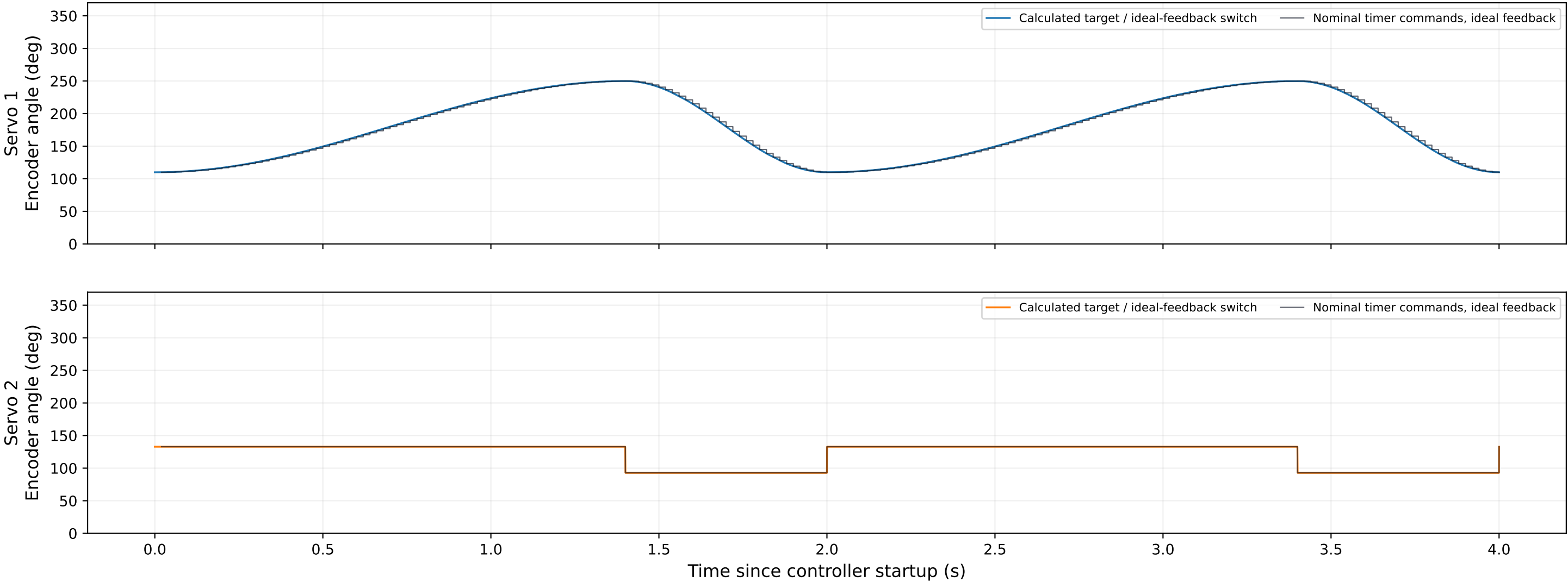


$u = (t \times 0.5) \bmod 1$; $b = 0.7$; $s = u/b$ during power, $(u-b)/(1-b)$ during recovery.
 $\theta_1 = -60 \cos(\pi s)$ during power; $+60 \cos(\pi s)$ during recovery.
 $\theta_2 = 45^\circ$ during power; 5° during recovery.
 $p_1 = \text{clip}(\text{int}(2048 + 4096\theta_1/360))$; $p_2 = \text{clip}(\text{int}(1000 + 4096\theta_2/360))$; clip to $[0, 4095]$.
Plotted encoder angle = $p \times 360/4096$ degrees. Above position equations retain raw counts for exact rounding.

Sin (LLU) | Square (LNR) / Run 7

f=0.5 Hz; S1 amplitude=70.00°; centers=180.00° / 87.89°; power=0.7

MODEL ONLY • Table settings + controller equations • No measured force or motion

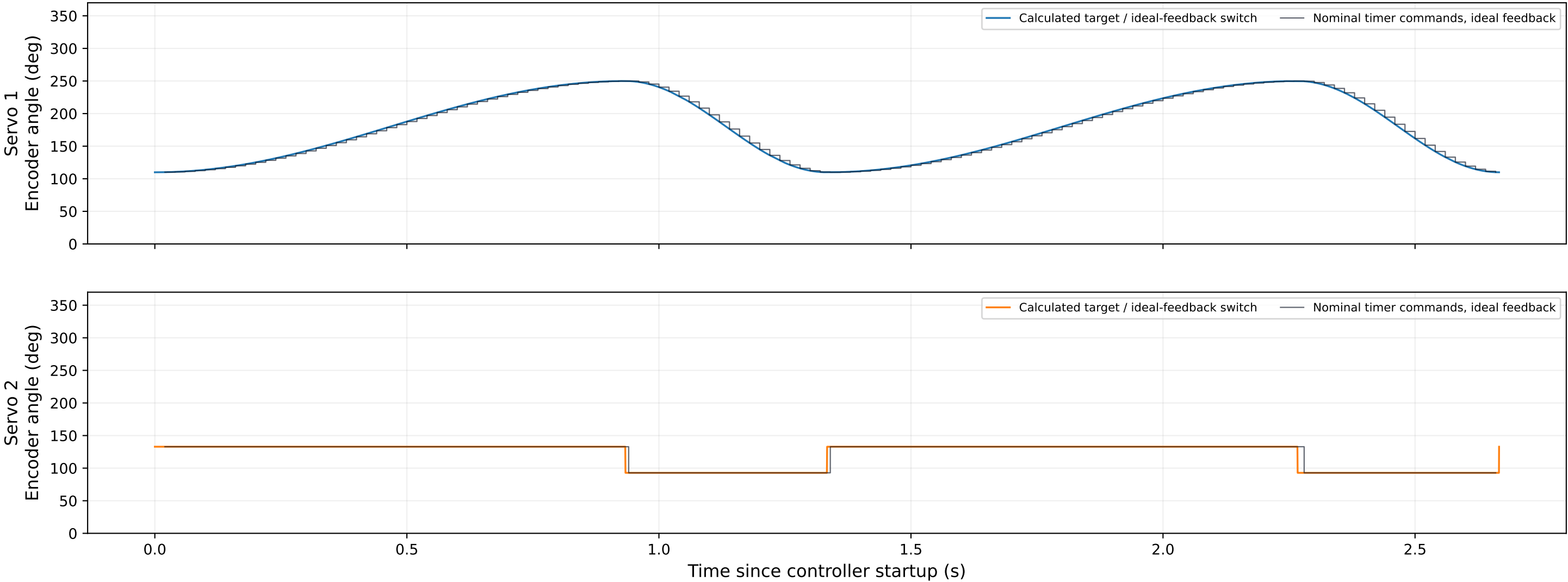


$u = (t \times 0.5) \bmod 1$; $b = 0.7$; $s = u/b$ during power, $(u-b)/(1-b)$ during recovery.
 $\theta_1 = -70 \cos(\pi s)$ during power; $+70 \cos(\pi s)$ during recovery.
 $\theta_2 = 45^\circ$ during power; 5° during recovery.
 $p_1 = \text{clip}(\text{int}(2048 + 4096\theta_1/360))$; $p_2 = \text{clip}(\text{int}(1000 + 4096\theta_2/360))$; clip to $[0, 4095]$.
Plotted encoder angle = $p \times 360/4096$ degrees. Above position equations retain raw counts for exact rounding.

Sin (LLU) | Square (LNR) / Run 8

f=0.75 Hz; S1 amplitude=70.00°; centers=180.00° / 87.89°; power=0.7

MODEL ONLY • Table settings + controller equations • No measured force or motion

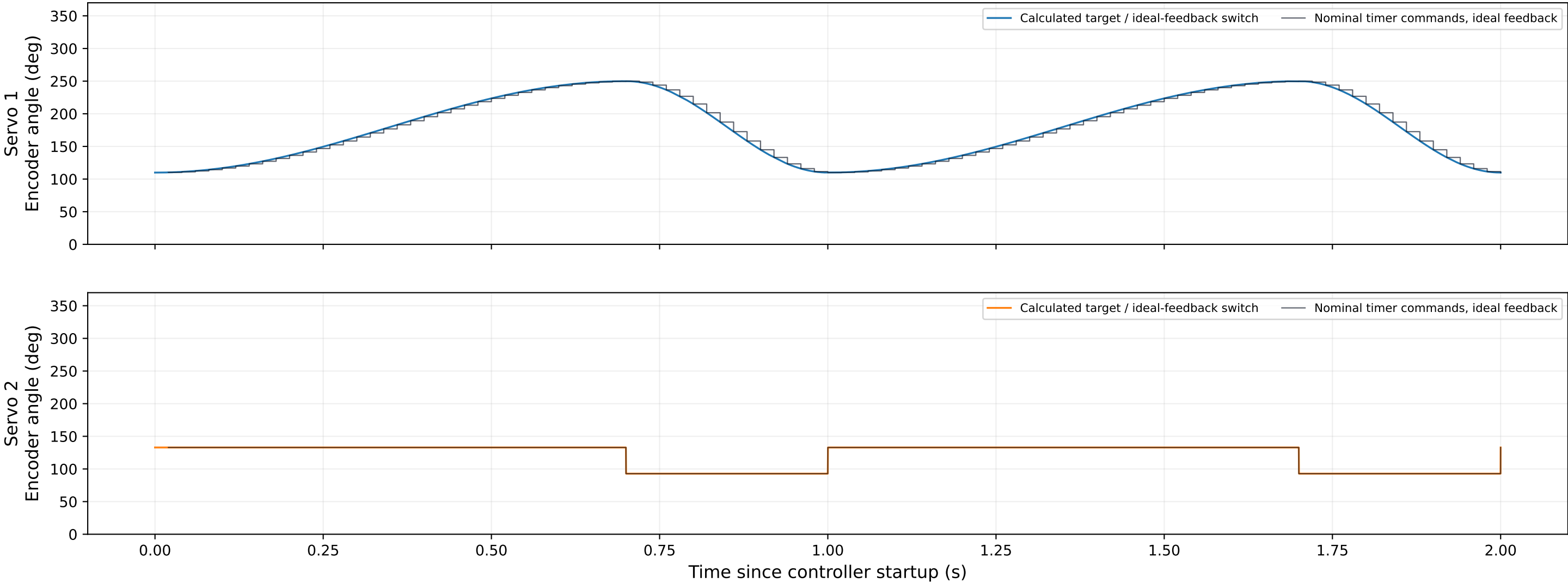


$u = (t \times 0.75) \bmod 1$; $b = 0.7$; $s = u/b$ during power, $(u-b)/(1-b)$ during recovery.
 $\theta_1 = -70 \cos(\pi s)$ during power; $+70 \cos(\pi s)$ during recovery.
 $\theta_2 = 45^\circ$ during power; 5° during recovery.
 $p_1 = \text{clip}(\text{int}(2048 + 4096\theta_1/360))$; $p_2 = \text{clip}(\text{int}(1000 + 4096\theta_2/360))$; clip to $[0, 4095]$.
Plotted encoder angle = $p \times 360/4096$ degrees. Above position equations retain raw counts for exact rounding.

Sin (LLU) | Square (LNR) / Run 9

f=1 Hz; S1 amplitude=70.00°; centers=180.00° / 87.89°; power=0.7

MODEL ONLY • Table settings + controller equations • No measured force or motion

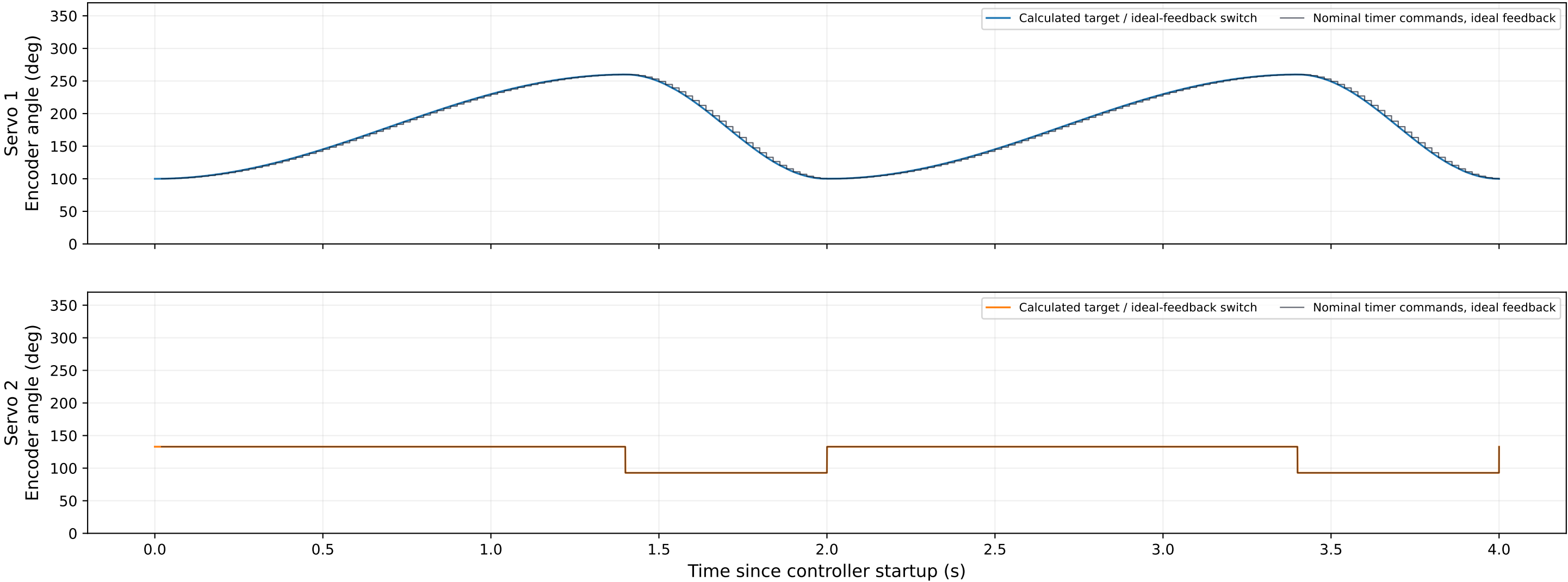


$u = (t \times 1) \bmod 1$; $b = 0.7$; $s = u/b$ during power, $(u-b)/(1-b)$ during recovery.
 $\theta_1 = -70 \cos(\pi s)$ during power; $+70 \cos(\pi s)$ during recovery.
 $\theta_2 = 45^\circ$ during power; 5° during recovery.
 $p_1 = \text{clip}(\text{int}(2048 + 4096\theta_1/360))$; $p_2 = \text{clip}(\text{int}(1000 + 4096\theta_2/360))$; clip to $[0, 4095]$.
Plotted encoder angle = $p \times 360/4096$ degrees. Above position equations retain raw counts for exact rounding.

Sin (LLU) | Square (LNR) / Run 10

f=0.5 Hz; S1 amplitude=80.00°; centers=180.00° / 87.89°; power=0.7

MODEL ONLY • Table settings + controller equations • No measured force or motion

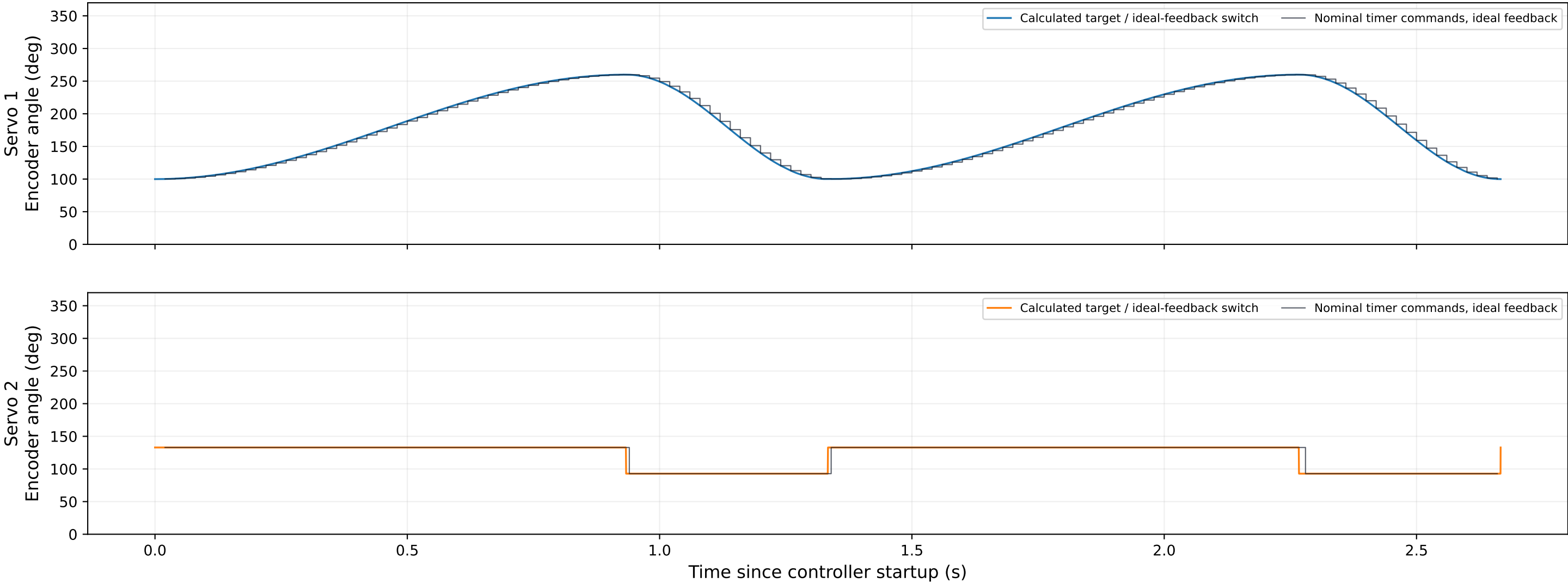


$u = (t \times 0.5) \bmod 1$; $b = 0.7$; $s = u/b$ during power, $(u-b)/(1-b)$ during recovery.
 $\theta_1 = -80 \cos(\pi s)$ during power; $+80 \cos(\pi s)$ during recovery.
 $\theta_2 = 45^\circ$ during power; 5° during recovery.
 $p_1 = \text{clip}(\text{int}(2048 + 4096\theta_1/360))$; $p_2 = \text{clip}(\text{int}(1000 + 4096\theta_2/360))$; clip to $[0, 4095]$.
Plotted encoder angle = $p \times 360/4096$ degrees. Above position equations retain raw counts for exact rounding.

Sin (LLU) | Square (LNR) / Run 11

f=0.75 Hz; S1 amplitude=80.00°; centers=180.00° / 87.89°; power=0.7

MODEL ONLY • Table settings + controller equations • No measured force or motion

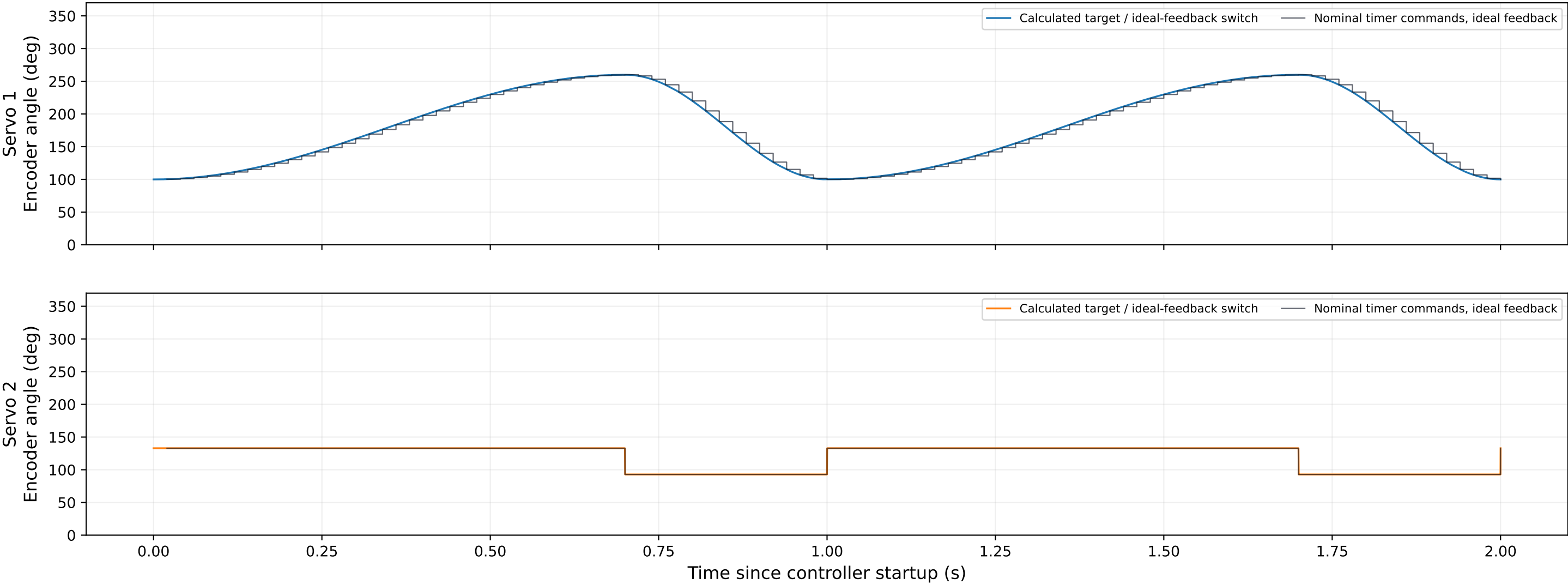


$u = (t \times 0.75) \bmod 1$; $b = 0.7$; $s = u/b$ during power, $(u-b)/(1-b)$ during recovery.
 $\theta_1 = -80 \cos(\pi s)$ during power; $+80 \cos(\pi s)$ during recovery.
 $\theta_2 = 45^\circ$ during power; 5° during recovery.
 $p_1 = \text{clip}(\text{int}(2048 + 4096\theta_1/360))$; $p_2 = \text{clip}(\text{int}(1000 + 4096\theta_2/360))$; clip to $[0, 4095]$.
Plotted encoder angle = $p \times 360/4096$ degrees. Above position equations retain raw counts for exact rounding.

Sin (LLU) | Square (LNR) / Run 12

f=1 Hz; S1 amplitude=80.00°; centers=180.00° / 87.89°; power=0.7

MODEL ONLY • Table settings + controller equations • No measured force or motion



$u = (t \times 1) \bmod 1$; $b = 0.7$; $s = u/b$ during power, $(u-b)/(1-b)$ during recovery.
 $\theta_1 = -80 \cos(\pi s)$ during power; $+80 \cos(\pi s)$ during recovery.
 $\theta_2 = 45^\circ$ during power; 5° during recovery.
 $p_1 = \text{clip}(\text{int}(2048 + 4096\theta_1/360))$; $p_2 = \text{clip}(\text{int}(1000 + 4096\theta_2/360))$; clip to $[0, 4095]$.
Plotted encoder angle = $p \times 360/4096$ degrees. Above position equations retain raw counts for exact rounding.

Sin (NU) | Square (R)

Original key: NESTEDSIN / PDF p.5 / rows map to run numbers

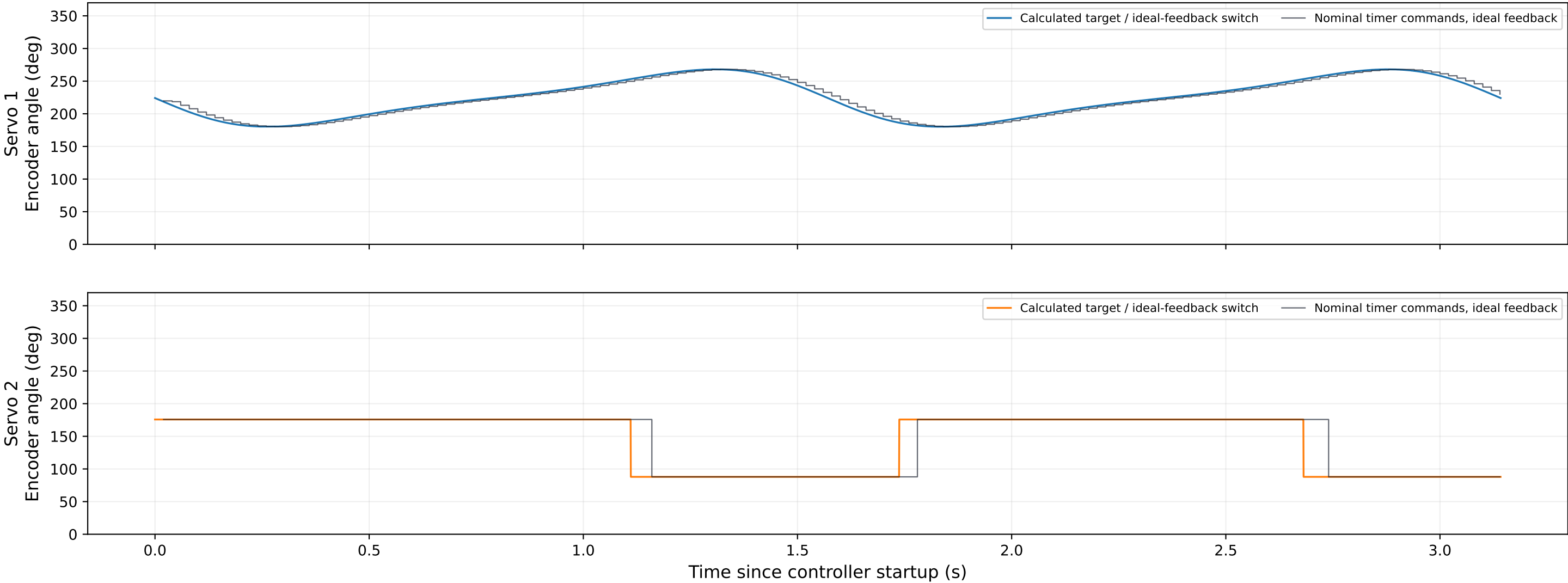
Run	f (Hz)	A1 (deg)	C1 (deg)	A2 / power (deg)	C2 (deg)	Direction	Power ratio	Extra
1	0.63662	43.95	224.12	87.9/175.8	175.78	-1	—	speed 4, omega 0.6
2	0.95493	43.95	219.73	87.9/175.8	175.78	1	—	speed 6, omega 0.7

All angles above are degrees. Centers use encoder zero; amplitudes are offsets. Harmonic amplitudes are scale factors.
Degrees = counts × 360/4096. Raw count parameters are retained in the CSV/JSON and exact position equations.

Sin (NU) | Square (R) / Run 1

f=0.63662 Hz; S1 amplitude=43.95°; centers=224.12° / 175.78°; direction=-1; speed=4; modulation=0.6

MODEL ONLY • Table settings + controller equations • No measured force or motion

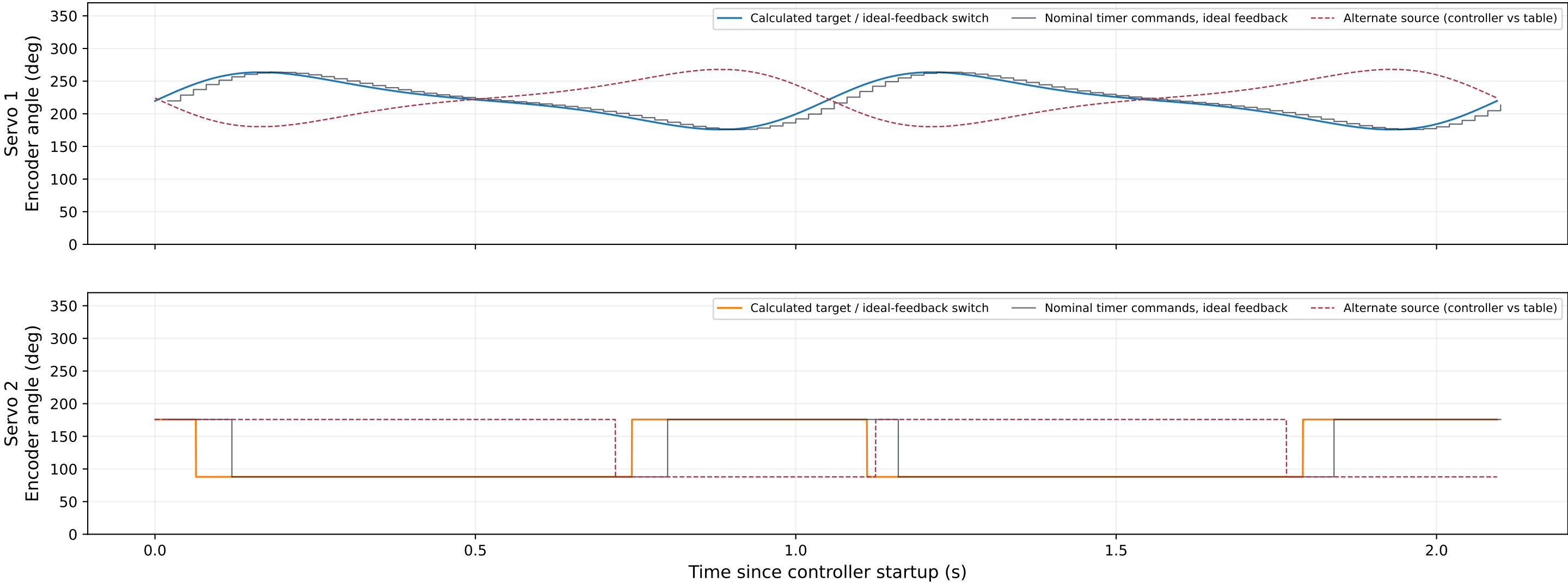


p1=2550+round(-500 sin(4t+0.6 sin(4t)))
q2[k+1]=1000 if e1[k]>2880; 2000 if e1[k]<2120; otherwise q2[k].
Plotted encoder angle = p × 360/4096 degrees. Above position equations retain raw counts for exact rounding.

Sin (NU) | Square (R) / Run 2

f=0.95493 Hz; S1 amplitude=43.95°; centers=219.73° / 175.78°; direction=1; speed=6; modulation=0.7

MODEL ONLY • Table settings + controller equations • No measured force or motion



p1=2500+round(500 sin(6t+0.7 sin(6t)))
q2[k+1]=1000 if e1[k]>2800; 2000 if e1[k]<2200; otherwise q2[k].
Plotted encoder angle = p × 360/4096 degrees. Above position equations retain raw counts for exact rounding.
Conflict: table center1=2500, direction=+1; pasted controller uses 2550, -1.

Sin (NU) (Slow/Fast) | Square (R)

Original key: SLOWFAST / PDF p.7 / rows map to run numbers

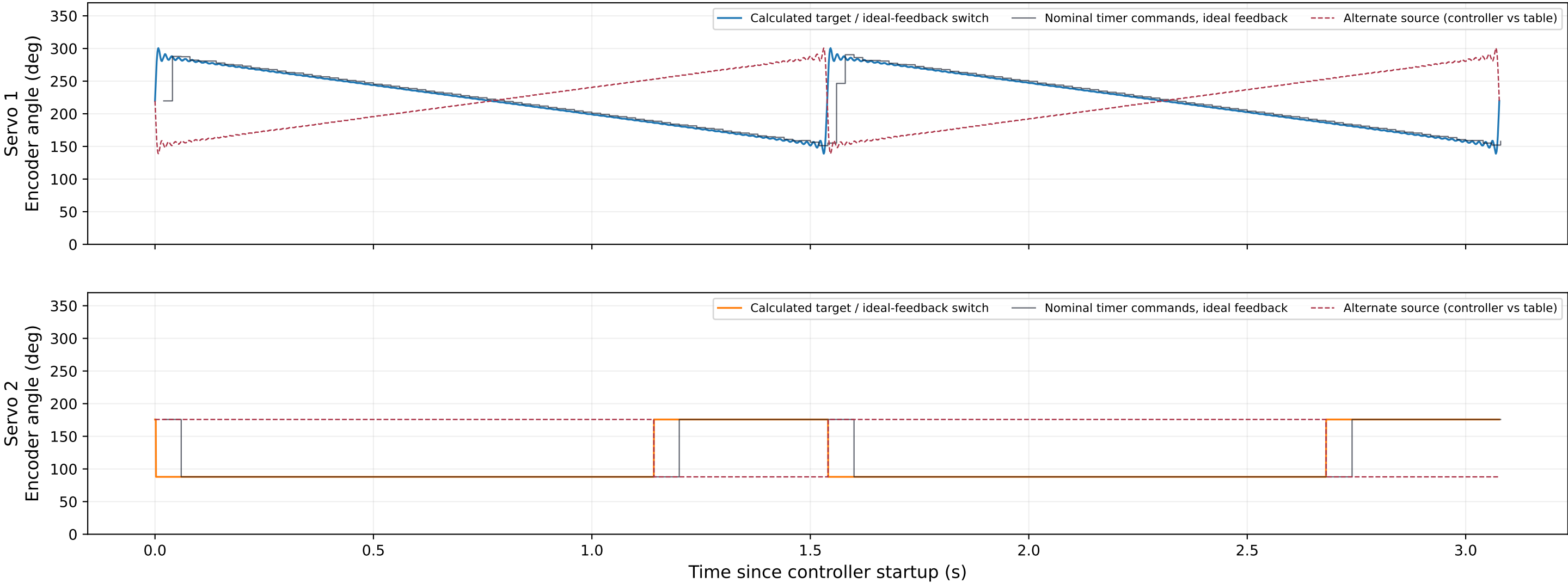
Run	f (Hz)	A1 (deg)	C1 (deg)	A2 / power (deg)	C2 (deg)	Direction	Power ratio	Extra
1	0.65	43.95	219.73	87.9/175.8	175.78	1	—	limits 186.3/253.1
2	0.85	43.95	219.73	87.9/175.8	175.78	1	—	limits 193.4/246.1

All angles above are degrees. Centers use encoder zero; amplitudes are offsets. Harmonic amplitudes are scale factors.
Degrees = counts × 360/4096. Raw count parameters are retained in the CSV/JSON and exact position equations.

Sin (NU) (Slow/Fast) | Square (R) / Run 1

f=0.65 Hz; S1 scale=43.95°; centers=219.73° / 175.78°; direction=1

MODEL ONLY • Table settings + controller equations • No measured force or motion

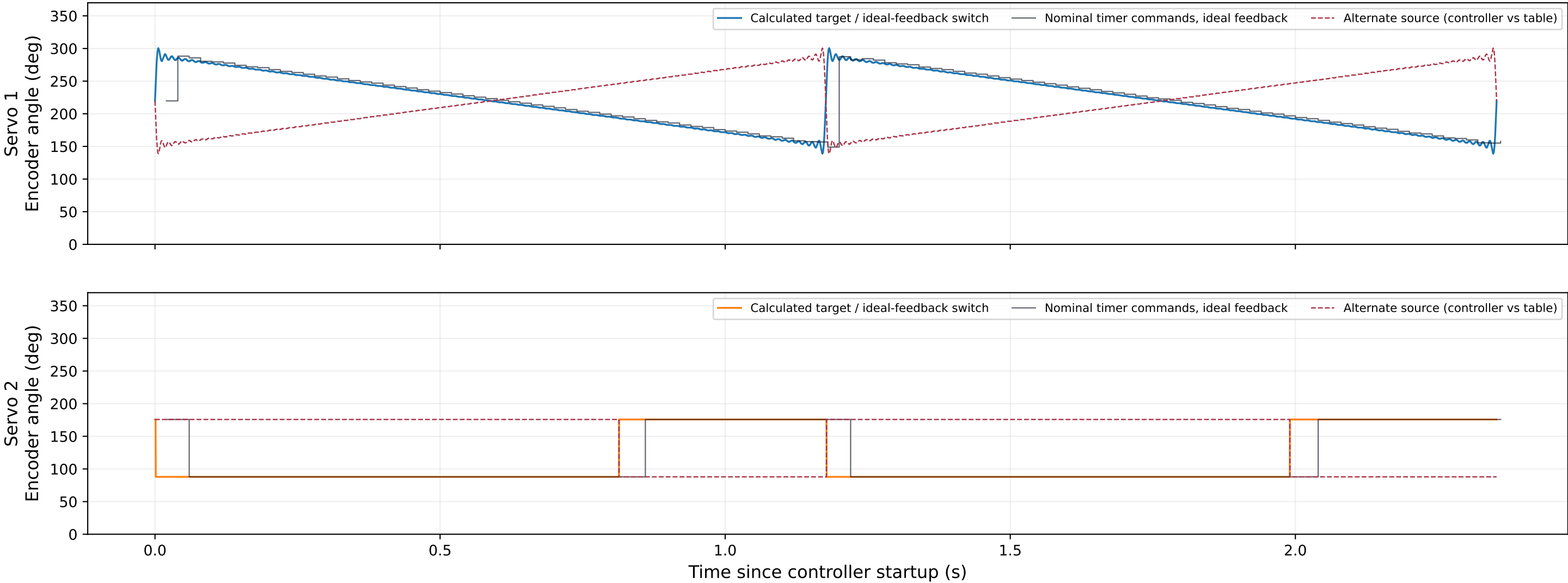


$p1 = 2500 + (1) \times 5 \cdot \text{round}([\sum_{n=1..99} 500 \sin(n \times 1.3\pi t)/n]/5)$
 $q2[k+1] = 1000$ if $e1[k] > 2880$; 2000 if $e1[k] < 2120$; otherwise $q2[k]$.
Plotted encoder angle = $p \times 360/4096$ degrees. Above position equations retain raw counts for exact rounding.
Conflict: table direction=+1; pasted controllers use -1. Formula image also uses -1.

Sin (NU) (Slow/Fast) | Square (R) / Run 2

f=0.85 Hz; S1 scale=43.95°; centers=219.73° / 175.78°; direction=1

MODEL ONLY • Table settings + controller equations • No measured force or motion



$p1=2500+(1)\times5 \text{ round}([\Sigma(n=1..99) 500 \sin(n\times1.7\pi t)/n]/5)$
 $q2[k+1]=1000 \text{ if } e1[k]>2800; 2000 \text{ if } e1[k]<2200; \text{ otherwise } q2[k].$
Plotted encoder angle = $p \times 360/4096$ degrees. Above position equations retain raw counts for exact rounding.
Conflict: table direction=+1; pasted controllers use -1. Formula image also uses -1.

Sin (U) | Fourier

Original key: SINFOURIER / PDF p.10 / rows map to run numbers

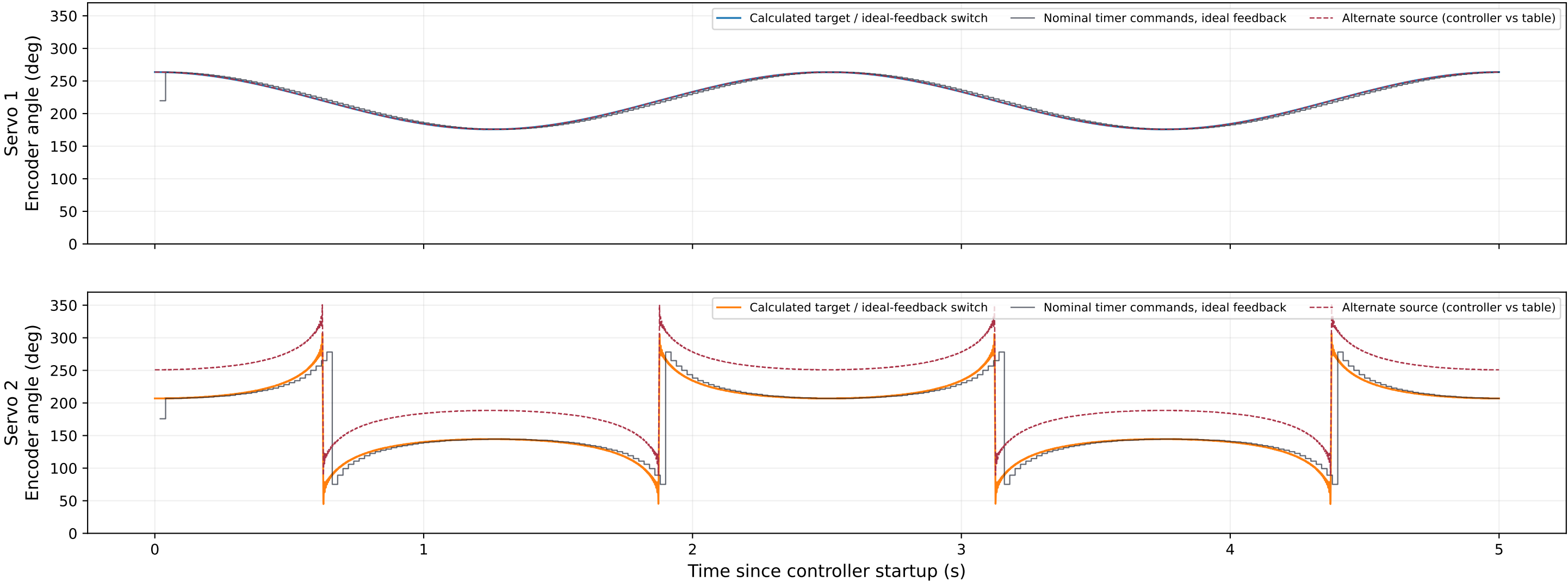
Run	f (Hz)	A1 (deg)	C1 (deg)	A2 / power (deg)	C2 (deg)	Direction	Power ratio	Extra
1	0.4	43.95	219.73	114.26	175.78	1	—	
2	0.65	43.95	219.73	114.26	175.78	1	—	

All angles above are degrees. Centers use encoder zero; amplitudes are offsets. Harmonic amplitudes are scale factors.
Degrees = counts × 360/4096. Raw count parameters are retained in the CSV/JSON and exact position equations.

Sin (U) | Fourier / Run 1

f=0.4 Hz; S1 amplitude=43.95°; centers=219.73° / 175.78°; direction=1

MODEL ONLY • Table settings + controller equations • No measured force or motion

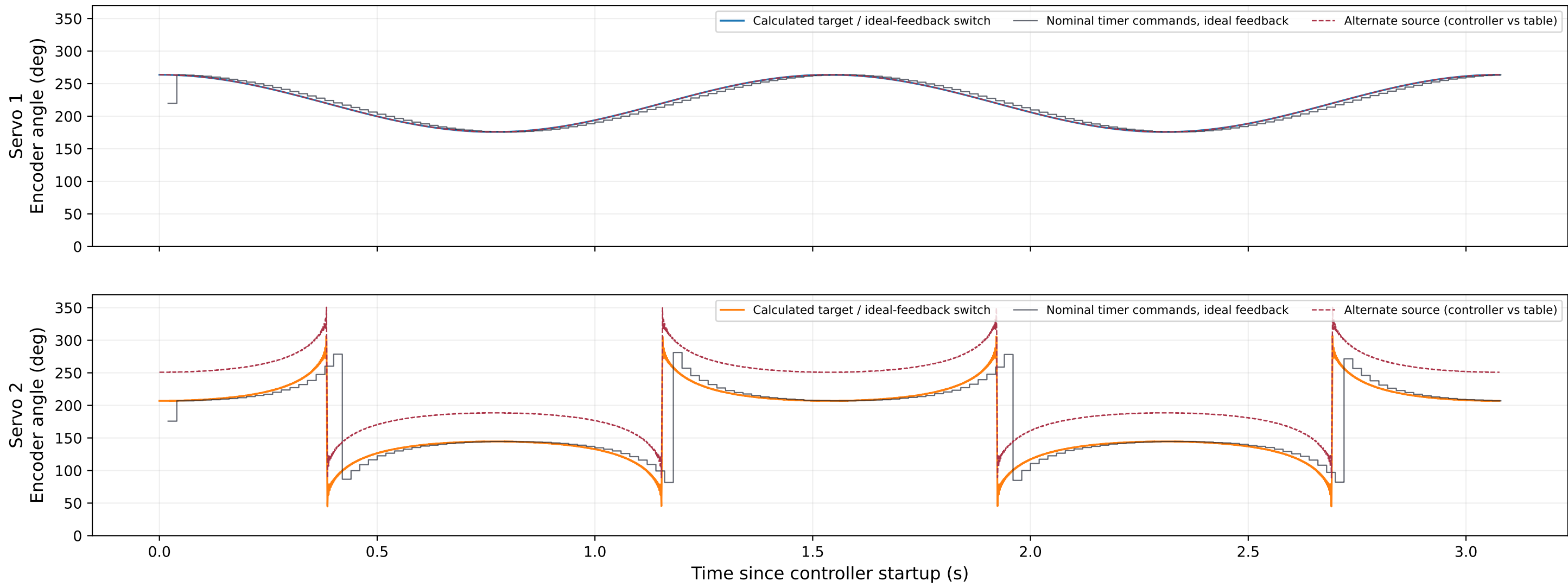


$p1 = \text{round}(2500 + 500 \cos(0.8\pi t))$
 $p2 = 2000 + 5 \cdot \text{round}(F(t)/5)$; $F = (4 \times 1300 / \pi) \sum_{n=1..299} (-1)^{(n+1)} \cos((2n-1) \times 0.8\pi t) / (2n+1)$
Plotted encoder angle = $p \times 360 / 4096$ degrees. Above position equations retain raw counts for exact rounding.
Conflict: table center2=2000; formula image and controller use 2500. Series scale 1300 comes from controller.

Sin (U) | Fourier / Run 2

f=0.65 Hz; S1 amplitude=43.95°; centers=219.73° / 175.78°; direction=1

MODEL ONLY • Table settings + controller equations • No measured force or motion



p1=round(2500+500 cos(1.3πt))
p2=2000+5 round(F(t)/5); F=(4×1300/π) Σ(n=1..299) (-1)^(n+1) cos((2n-1)×1.3πt)/(2n+1)
Plotted encoder angle = p × 360/4096 degrees. Above position equations retain raw counts for exact rounding.
Conflict: table center2=2000; formula image and controller use 2500. Series scale 1300 comes from controller.

Fourier | Square (R)

Original key: FOURIER / PDF p.11 / rows map to run numbers

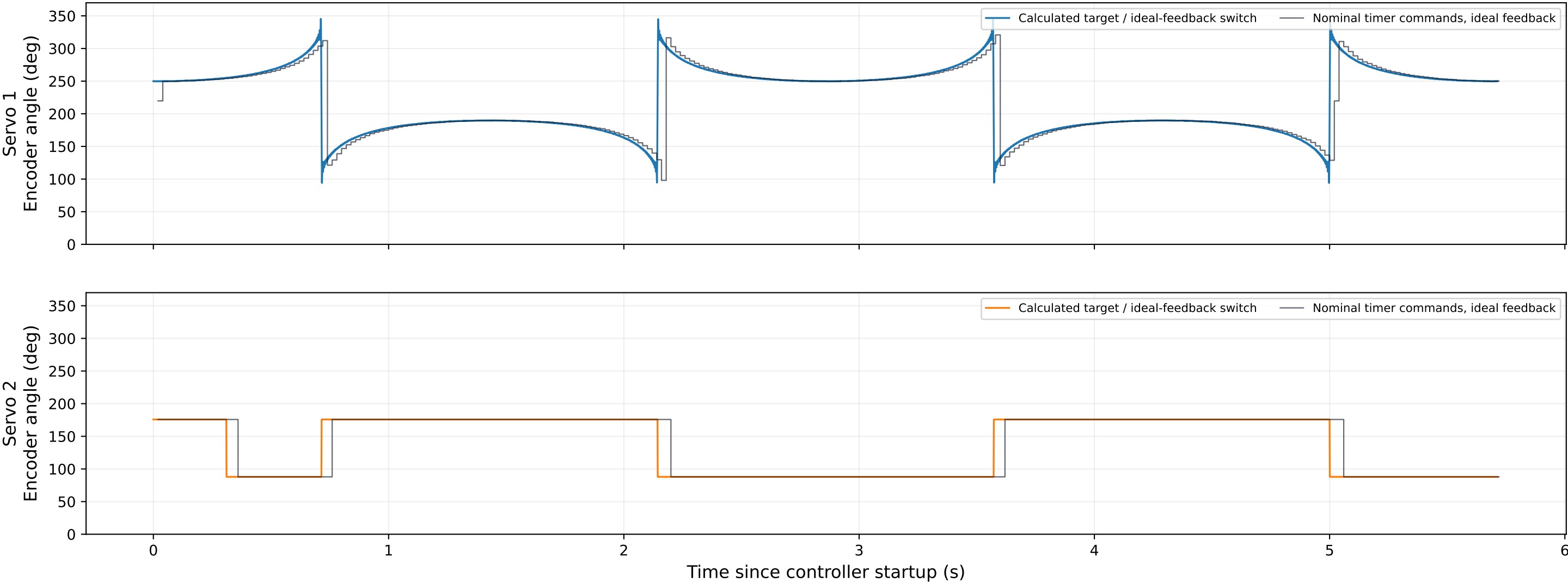
Run	f (Hz)	A1 (deg)	C1 (deg)	A2 / power (deg)	C2 (deg)	Direction	Power ratio	Extra
1	0.35	109.86	219.73	87.9/175.8	175.78	1	—	limits 184.6/254.9
2	0.65	109.86	219.73	87.9/175.8	175.78	1	—	limits 184.6/254.9

All angles above are degrees. Centers use encoder zero; amplitudes are offsets. Harmonic amplitudes are scale factors.
Degrees = counts × 360/4096. Raw count parameters are retained in the CSV/JSON and exact position equations.

Fourier | Square (R) / Run 1

f=0.35 Hz; S1 scale=109.86°; centers=219.73° / 175.78°; direction=1

MODEL ONLY • Table settings + controller equations • No measured force or motion

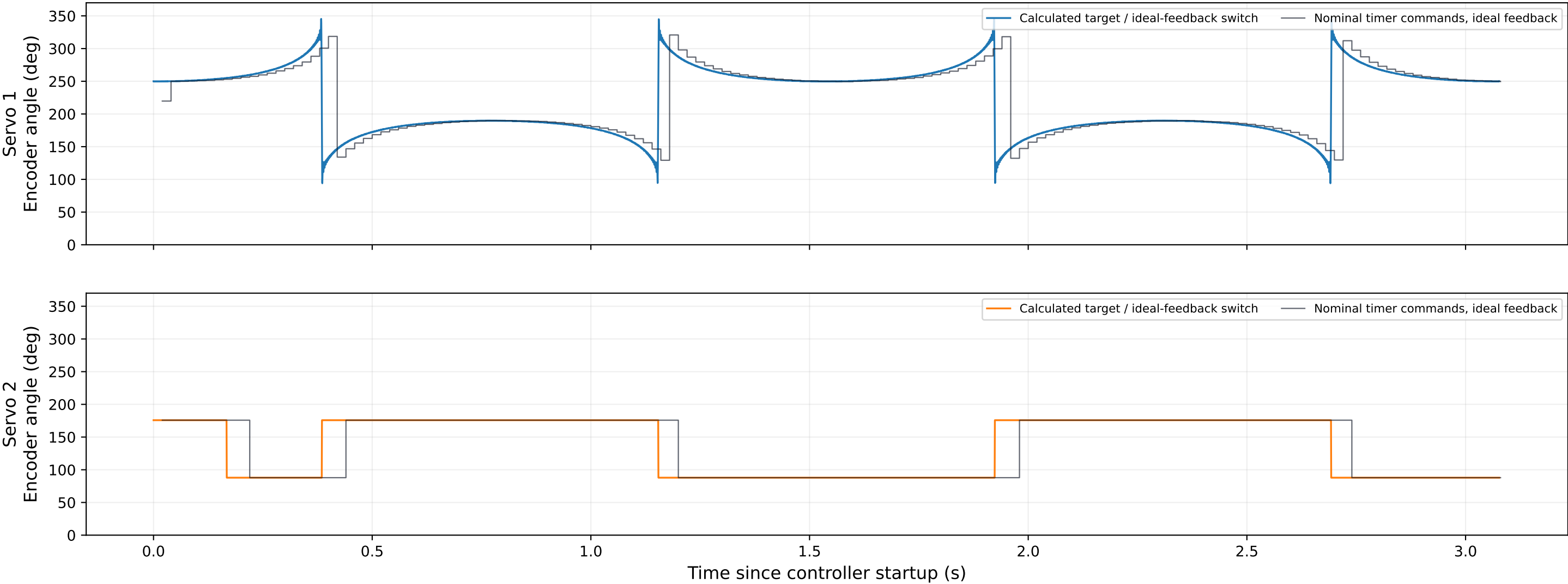


$p1=2500+5 \text{ round}(F(t)/5)$; $F=(4 \times 1250/\pi) \sum_{n=1..299} (-1)^{(n+1)} \cos((2n-1) \times 0.7\pi t)/(2n+1)$
 $q2[k+1]=1000$ if $e1[k]>2900$; 2000 if $e1[k]<2100$; otherwise $q2[k]$.
Plotted encoder angle = $p \times 360/4096$ degrees. Above position equations retain raw counts for exact rounding.

Fourier | Square (R) / Run 2

f=0.65 Hz; S1 scale=109.86°; centers=219.73° / 175.78°; direction=1

MODEL ONLY • Table settings + controller equations • No measured force or motion



$p1=2500+5 \text{ round}(F(t)/5)$; $F=(4 \times 1250/\pi) \sum_{n=1..299} (-1)^{(n+1)} \cos((2n-1) \times 1.3\pi t)/(2n+1)$
 $q2[k+1]=1000$ if $e1[k]>2900$; 2000 if $e1[k]<2100$; otherwise $q2[k]$.
Plotted encoder angle = $p \times 360/4096$ degrees. Above position equations retain raw counts for exact rounding.